

Evaluating the Environmental Impact: A Multi-Sensor Remote Sensing Approach for Spatial and Temporal Analysis

Bin Zhu, Ahsen Maqsoom, Lapyote Prasittisopin, Chaudhary Danyal Aslam, Umer Khalil, Sahar Zia, Niamat Ullah, M. Abdullah-Al-Wadud, Rebouh Nazih Yacer

Abstract— This study quantifies the potential environmental impacts of the Karakoram Highway (KKH) construction, which links the northern region of Pakistan with Western China. The upgrade of the KKH was carried out under the China-Pakistan Economic Corridor (CPEC) project. We examined highway construction's spatial and temporal effects on the immediate environment and the ecological revival progressions through remotely sensed images taken at distinct points in time. Here, using a buffer zone of 20 km along the KKH (10 km on both sides), we estimated the before-during-after remote sensing-based factors that relate to the ecology to compute the geographical and temporal variations of environmental effects during the building of the KKH. The outcomes showed that whereas land cover makeup remained broadly consistent in the south of the buffer, it underwent significant changes in the upper portion as we moved more and more towards the north. The buffer region showed clear degradation-recovery trends in the moistness and vegetation states, as evidenced by the Normalized Difference Moisture Index (NDMI) and the Normalized Difference Vegetation Index (NDVI) correspondingly. Meanwhile, the Land Surface Temperature (LST) gradually increased. The maximum relative changes in NDMI, NDVI, and LST were roughly 60%, 40%, and 12%, respectively. According to an Integrated Environment Quality Index (EQI), the effects of highway developments on the environment were most pronounced in the immediate vicinity and diminished with distance. This study's method for quantifying highway system disturbances' spatial and temporal changes and subsequent recovery can be easily extended to different geographical areas.

Index Terms— Karakoram Highway; China-Pakistan Economic Corridor; Highway Construction; Ecological Quality Index; Satellite Indices

This paper has been supported by the RUDN University Strategic Academic Leadership Program (Corresponding author: Umer Khalil)

Bin Zhu is with the Department of Fine Arts and Design, Leshan Normal University, Leshan, Sichuan, 614000, China (e-mail: zhubin@lsnu.edu.cn).

Ahsen Maqsoom is with the Green Tech Institute, University Mohammed VI Polytechnic, Morocco, and the Faculty of Architecture, Chulalongkorn University, Thailand (e-mail: Ahsen.maqsoom@um6p.ma),

Chaudhary Danyal Aslam is with the Department of Civil Engineering, COMSATS University Islamabad Wah Campus, Pakistan (Chdanyal99@gmail.com).

Lapyote Prasittisopin is with the Faculty of Architecture, Chulalongkorn University, Thailand (lapyote.p@chula.ac.th).

Umer Khalil is with the ITC Faculty of Geo-Information Science and Earth Observation, University of Twente, 7522 NB Enschede, The Netherlands (e-mail: umerkhalil745@gmail.com).

Niamat Ullah is with the Spatial Decision Support System (SDSS) Lab, National Center of GIS and Space Applications (NCGSA), Balochistan University of Information Technology, Engineering and Management Sciences (BUIEMS), Quetta, 87300, Pakistan (e-mail: niamatullahza@gmail.com).

Sahar Zia is with the Geography Department, Lahore College for Women University (LCWU), Lahore, Pakistan (Email: sahar.zia@lcwu.edu.pk)

M. Abdullah-Al-Wadud is with the Department of Software Engineering, College of Computer and Information Sciences, King Saud University, Riyadh 11543, Saudi Arabia (e-mail: mwadud@ksu.edu.sa).

Rebouh Nazih Yacer is with the Department of Environmental Management, Institute of Environmental Engineering, RUDN University,

Miklukho-Maklaya St., 117198 Moscow, Russia (e-mail: n.yacer16@outlook.fr).

I. INTRODUCTION

ALTHOUGH the Karakoram Highway (KKH) road network, which is a pathway of the China-Pakistan Economic Corridor (CPEC), provides essential economic and social facilities to individual communities, incorporating trade exchange, transport facilities, ethnic interchange, information, and the movement of materials [1], [2]. At the same time, they also cause various detrimental ecological effects (such as habitat loss, landscape fragmentation, species extinction, soil disturbance, and ecosystem degradation) due to edge effects, acoustic disruptions, landscape segmentation, and human-aided disease dispersal [3]. The extensive construction of highways dramatically affects the environment and its structure [4], [5].

Additionally, they impact the local climate and heat transfer in the vicinity [6], [7]. The amount of solar radiation energy used for plant evapotranspiration reduces after vegetation removal due to highway construction, increasing the landscape's surface temperature and changing the region's climate [8]. Because of the sizeable existent road networks and the anticipated significant increase in road length in emerging countries, it is more important than ever to comprehend the temporal and spatial distribution of these environmental consequences induced by highways [9].

Multiple studies have shown that highway construction and maintenance can affect landscape and ecology [10]. It has been common practice to enumerate the ecological consequences of road building using the road-effect or effect distance zone [11], the space over which considerable ecological consequences reach outwards from a path [12]. For transportation planning and maintenance activities, it is crucial to identify the road-effect zone [13].

Additionally, the placement of roads in a landscape is heavily influenced by road patterns, which are the results of human activity and land-use patterns. There are various effect distances or road-effect zones due to highway construction, including the destruction of habitats [14], extinction of species [15], fragmentation of the landscape, and degradation of the ecosystem [16]. Therefore, detecting and evaluating land-use changes and their effects on habitat quality must come before highway construction, especially inside road-effect zones [17], [18].

Different authors realized that different effect distances coexist because the effect distance is variable dependent. Because of wind, slope, and environmental changes on either

side of the road, the effect distances are also asymmetric concerning a road [19]. The road-effect zone also varies longitudinally along the road and is symmetric to the road [19], [20]. Estimating the part of the land that is environmentally impacted by roads and identifying the essential portions or sections for restoration and conservation are made possible by marking the longitudinal alterations of a road-effect zone [20], [21]. Regardless of the value of the road-effect zone, it remains quite challenging to locate it. This is primarily due to the requirement for recurring sampling in space and time, which is costly and challenging [22].

Using the Normalized Difference Vegetation Index (NDVI) derived from Landsat, Gulci et al. [23] discovered that between 2000 and 2015, over 18% of the coniferous forests inside a 100-m zone alongside the road were ruined. Using NDVI obtained from Landsat satellite images, Alphan and Derse [24] examined landscape changes in the Mediterranean and discovered that woodland loss appeared at small elevations adjacent to highways. It is important to note that the majority of these findings only use limited images that were cloud-free every few years, which may not be enough to portray the temporal alterations in road influences from building to operation, let alone reduce the oversights brought on by images taken at various time intervals [25], [26]. Therefore, New methods based on remote sensing must be created to facilitate monitoring and evaluation of highways' environmental effects.

This paper follows the study of Alphan et al. [24], using a unique technique of remote sensing images obtained at different time intervals to assess the geographical and temporal effects of the Karakoram Highway (KKH) road network development on the immediate environment and the progression of ecological revival in the northern region of Pakistan. Provincial land use planners, policy-makers, road engineers, researchers, and many other specialists in Pakistan who manage resources will find this material helpful to understand how a roadway affects landscape patterns and ecosystem processes to help restore disturbed landscapes and reduce road networks' effect on the environment [27].

Feng et al. [28] developed a BDACI-RS methodology to measure the effect distance of a roadway through remotely sensed data. It showed that it is impossible to accurately measure the road-effect distance utilizing the alterations in median and mean circumstances, as achieved in several other analyses. The spatial modifications of the indicators showed that the KKH's influence distance was at least 1000 m from the highway. Sizable Land Use Land Cover (LULC) changes were recorded in the areas adjacent to the highway. The LST gradually increased, while the vegetation and wetness conditions in the buffer zone demonstrated decrease-increase (degradation-recovery) trends that aligned effectively with the development and operation phases of the roadway. The NDVI and the NDMI showed these trends [29].

Using Environment Impact Assessment (EIA), Khan [30] concentrated on how roads affect ecosystems and the parts of ecosystems, habitats, and species that comprise ecosystems, incorporating stream sediments, water quality, and riparian processes. They also looked into how roads affected species and

ecosystems on land and water. They stated that the environmental consequences of CPEC might incorporate temporal and spatial dimensions along a highway and abiotic and biotic constituents associated with an enormous road network. Additionally, supplementary implicit influences on ecosystems and habitats are triggered by the footprint of certain connecting roads. Khan [30] describes that road networks change light states on motorways and disturb soil conditions under immense development or renovation. The spatial consequences of this project may alter because of species habitat constraints and ecosystem attributes [17].

Temporal negative externalities related to roads can occur due to road development, road use, and maintenance of the roads. An earlier study by Soanes et al. [31] has underlined that roads might adversely affect landscape levels, habitats, and species. Another study by Rehman et al. [26] summarised the road effects of the Pindi Bhattian Motorway construction, which can be observed because of the development of CPEC. As highlighted in the study, they can be found along a newly completed road and have long-term impacts. Khan et al. [30] looks at how road networks affect the lighting on highways and how they affect the state of the soil during major rehabilitation or construction. Due to ecological variables and species habitat requirements, this project's spatial effects could differ. Their development, use, and upkeep may bring temporary negative externalities associated with roads. A critical study by Gucinski et al. [32] shows that highways may disturb the extent of landscapes, habitats, and species. This effect can be frequent during the development of CPEC [33].

Nedbal and Brom et al. [34] calculated heat balance and surface functional parameters using multispectral Landsat satellite images and meteorological measurements gathered from the field before and during highway construction [35]. The findings revealed that constructing multi-lane highways can significantly influence the local climate conditions of a particular area [36]. Depending on the surrounding surface features, a surge in surface temperature of up to 7 °C will affect the actual highway site and the area within 45 to 75 m of the highway axis. This will decrease evapotranspiration by an average of 32.8-43.7 m³ each summer day for each kilometer of a highway and its vicinity. The latent heat flux will also be reduced by an average of 22.3-29.7 MWh per day per kilometer of a road and its vicinity, indicating a decline in the cooling effect of vegetation [37].

Furthermore, the realization of the ecological processes temporally is crucial for environmental monitoring, engineering, management, and mitigation [38]. Due to the vital significance of these concerns, planners and decision-makers require new information and scientific approaches to more effectively reduce the effects of highway development and support landscape restoration [39], [40]. As a result, an extensive time series of remote sensing data has significant importance and role because it can record distinct aspects of the ecosystem changes triggered by highway construction [41]. The specific goals of this study are to (1) develop a new general method for quantifying the environmental effects of highways using remote sensing data collected at various points in time for

change detection; (2) to investigate the effect distance for different ecological indicators alongside recently renovated KKH; and (3) establish a helpful framework for evaluating the quality of the environment.

II. MATERIALS AND METHODS

A. Study area

The study site is the Karakoram Highway (KKH), National Highway 35, or China -Pakistan Friendship Highway. It is a 1300 km highway section that extends from Hasan Abdal in the Punjab province of Pakistan to Bara Khun in Gilgit-Baltistan [42], [43]. It crosses into China at Khunjerab Pass in Gilgit-Baltistan and becomes China National Highway 314 (Figure 1), thus connecting Gilgit-Baltistan with Western China. Construction on the KKH began in 1962 and was finished by the end of 1978. Because of its irregular shape, the study area has been divided into two small portions, which are represented by Figure 1a and Figure 1b. Figure 1a represents the northern part of the highway, and Figure 1b represents the southern part.

This division is followed throughout the analysis, including all of the figures produced as part of the analysis.

The highway connects the northern parts of Pakistan to the rest of the country, particularly to the provinces of Punjab and Khyber-Pakhtunkhwa. A recent joint development initiative of China and Pakistan, also known as the CPEC, involved the upgrade and renovation of the KKH to support regional connectivity, thus boosting business and economy [28], [42], [44].

The highway is a popular tourist attraction and one of the world's highest paved roads, passing through the mountainous areas of Karakoram, Hindukush, and the Himalayan Mountain range that joins the Indian and Eurasian tectonic plates, including the Kohistan Island Arc [45]. The surrounding area along the KKH is a hilly landform with rough topography due to the presence of different mountain ranges, and the highest point along the highway is at a maximum elevation of around 6500 m near Khunjerab Pass. Most of the land in the immediate surrounding areas comprises cities, agricultural land, forests, water bodies, and small villages [44].

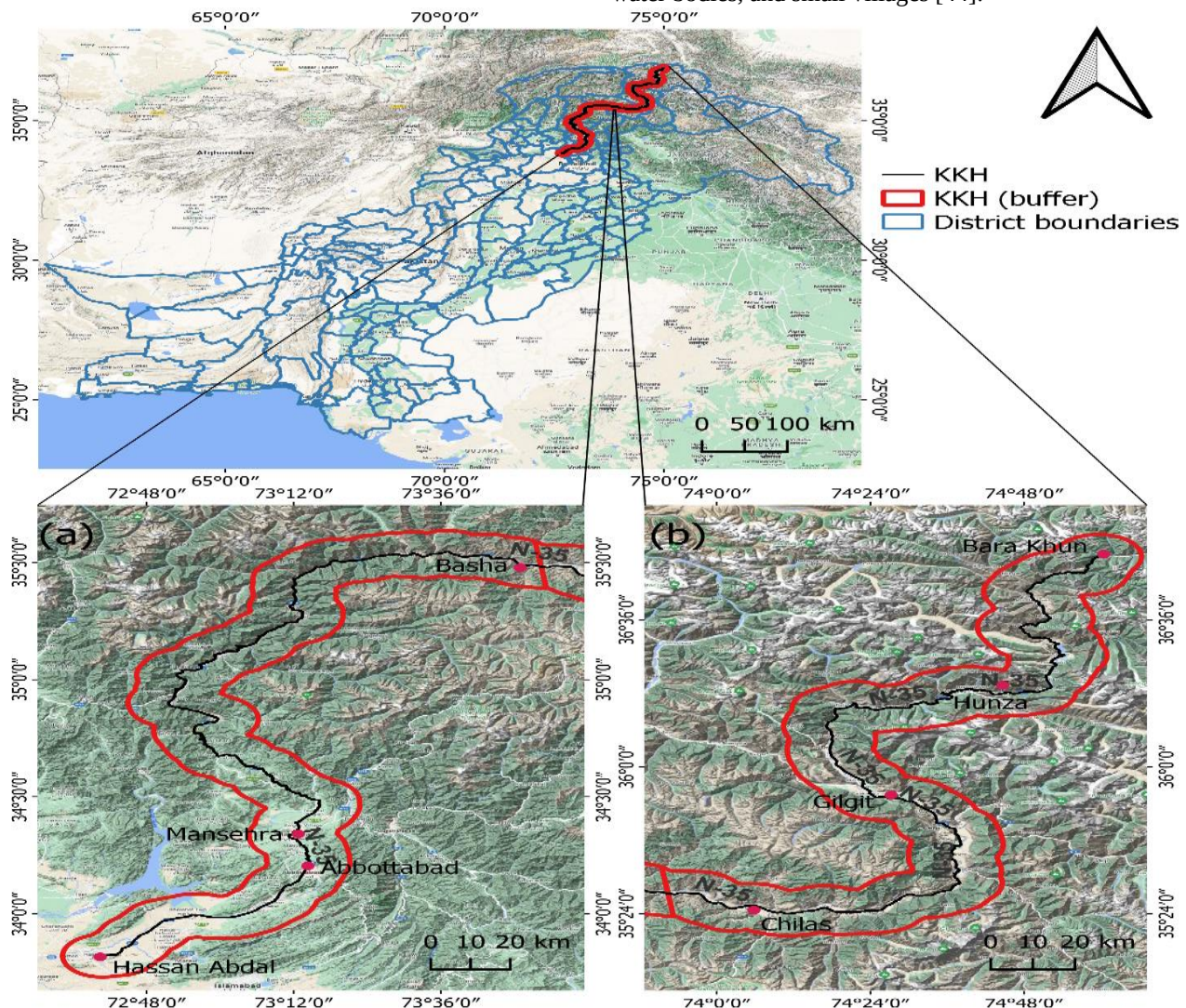


Figure 1. The Karakoram Highway (KKH) and the study area's location (figure represented with alphabet (a) represents the northern part of the highway, and the figure represented with alphabet (b) represents the southern part of the highway).

B. Remote sensing data

To map the changes in the environmental and ecological factors in terms of their spatial and temporal variation before, during, and after the construction of the KKH, we utilized several Landsat 8 satellite images acquired from 2013 to 2022. These images were downloaded from a commonly used open-access source, the USGS Earth Explorer (<https://earthexplorer.usgs.gov/>, accessed on June 10, 2023) (Table I). The Landsat 8 images come with a 30 m spatial resolution and were collected primarily between May and July from 2013 to 2022 to extract the different ecological indicators. The slope of the study area was obtained from the 30 m spatial

resolution Digital Elevation Model (DEM) of the Shuttle Radar Topography Mission (SRTM). The DEM was also retrieved from the USGS Earth Explorer.

C. Calculation of Ecological Indicators

To illustrate different aspects of the ecological variations alongside the KKH, the subsequent three indicators were computed from Landsat images: (1) Normalized Difference Vegetation Index (NDVI), (2) Normalized Difference Moisture Index (NDMI), (3) Land Surface Temperature (LST). Below is a description of the processes used to derive these data layers.

TABLE I
DATA FROM LANDSAT WERE USED IN THIS INVESTIGATION.

Landsat	Sensor	Path	Row	Acquisition Date
LC8	OLI/TIRS	149	34	19-Jun-13
		149	35	19-Jun-13
		150	35	10-Jun-13
		150	36	10-Jun-13
		150	37	10-Jun-13
		149	34	29-Jul-16
		149	35	29-Jul-16
		150	35	20-Jul-16
		150	36	20-Jul-16
		150	37	20-Jul-16
		149	34	26-Aug-19
		149	35	23-Aug-19
		150	35	13-Jul-19
		150	36	13-Jul-19
		150	37	13-Jul-19
		149	34	31-Aug-22
		149	35	31-Aug-22
LC9	OLI-2 / TIRS-2	150	35	30-Aug-22
		150	36	30-Aug-22
		150	37	30-Aug-22

1) Normalized Difference Vegetate Index (NDVI)

In numerous earlier research [46], [47], the normalized vegetation index produced from satellite data was often used to evaluate vegetation responses to environmental change. It is one of the vegetation indices frequently used to analyze ecosystem dynamics and processes [46], [48]. This is how the NDVI is determined using Equation 1:

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)} \quad (1)$$

Where Red and NIR are the spectral reflectance measurements made in the Red and Near-Infrared wavelength ranges, correspondingly, higher index values are linked to increasing green leaf areas and biomass, and the index values for vegetation are primarily positive [49].

2) Normalized Difference Moisture Index (NDMI)

Canopy water content is a crucial parameter for assessing the stress and health of an ecosystem [46]. The NIR and Short-Wave Infrared (SWIR) channels of Landsat can be used to construct the normalized differential moisture index Equation 2 [50] as follows:

$$NDMI = \frac{(NIR - SWIR)}{(NIR + SWIR)} \quad (2)$$

The NDMI and canopy water content have a strong relationship [46], [50]. NDMI values range from 1 to 1, with higher values denoting more moisture [51].

3) Land Surface Temperature (LST)

Data from remote sensing can be used to determine the LST, a crucial indicator of ecological conditions [52], [53]. According to LST research, variable surface soil water content and vegetation cover affect how sensible and latent heat fluxes are distributed, affecting how the surface radiant temperature

responds [54], [55]. In this study, the radiative transfer equation (RTE) was utilized to calculate the LST using data from the Thermal Infrared Sensor (TIRS) infrared band 10 of Landsat 8 [56].

Following some previous studies, the LST was derived using a three-step process [56], [57], [58]. First, at-sensor spectral radiance was transformed from Digital Numbers (DN) for the

thermal bands in Landsat 8 (bands 10 and 11). Second, each pixel's Brightness Temperature (BT) was determined. Lastly, the at-sensor brightness temperature was changed to the LST via the variable spectral emissivity based on actual object attributes to reproduce the temperature changes among land cover types on the surface.

TABLE II
THE EVALUATION CRITERIA FOR ENVIRONMENTAL QUALITY AND THEIR WEIGHTS.

Factor Ecological	Indicator	Weight
Vegetation	NDVI	0.3
Moisture	NDMI	0.3
Surface temperature	LST	0.2
Topography	Slope	0.1
Elevation	DEM	0.1

D. Land Use and Land Cover (LULC)

LULC change, primarily the change from natural lands to impervious surfaces, is one of the significant effects of highway construction [59], [60]. Following the work by Karan et al [50], supervised classification was used to categorize LULC types alongside the KKH to investigate the effects of highway construction on LULC change, using Google Earth Engine (GEE) (<https://earthengine.google.com>, accessed on May 20, 2023). Along the KKH, four broadly defined LULC types were categorized: urban, vegetation, water, bare, and snow. Two hundred ground points from Google Earth Pro were chosen along the KKH as training samples to establish the classification criteria[61], [62]. These points included urban (40 points), water (30 points), vegetation (50 points), bare (50 points), and snow (20 points). To verify the accuracy of the classification, an additional 200 points for each LULC category were chosen throughout the zone along the highway.

E. Recognition of the Road Effect Zone

The construction of highways typically impacts the area immediately surrounding it, and the impact lessens as distance increases [63]. The five data layers utilized for ecological assessment were the NDVI, NDMI, DEM, LST, and slope. These different data layers were also checked for multicollinearity. It was discovered that no multicollinearity was present among the considered data layers for the analysis. Using ArcGIS 10.2, we looked at changes in environmental indicators within the buffer defined on both sides of the roadway to compute the effect distance of the KKH. The indicators were plotted to illustrate the zone-specific and location-specific influences. A timeline from 2013 to 2022 and location-specific impacts inside the used buffer were examined. These findings would explain the decline and improvement of environmental conditions near KKH before, during, and after highway construction.

F. Criteria for Layer Normalization

In most MCA approaches, normalizing the criteria components is an essential step. The six elements of our multi-Where w_j and $N_{j,i}$, i , are the weight and normalized value of indicator j for pixel i , respectively. Surveys of stakeholders

criteria analysis method have various dimensional properties. The normalization approach is implied to convert the evaluation criteria to similar units. Multiple techniques for normalizing evaluation criteria exist, including the linear max-min dimensionless, linear ratio-based, and sun-based dimensionless methods. The linear max-min dimensionless method is the most frequently employed method to standardize our study's criteria units. Equations (3) and (4) express it.

$$r_{ij} = \frac{X_{ij} - X_j^{min}}{X_j^{max} - X_j^{min}} \quad (3)$$

$$r_{ij} = \frac{X_j^{max} - X_{ij}}{X_j^{max} - X_j^{min}} \quad (4)$$

Where r_{ij} represents the normalized values for the i th score of the j criterion layer, X_{ij} represents the i th score's raw value, and X_j^{max} and X_j^{min} stand for the layer's maximum and minimum scores, respectively. The equation to be used will depend on whether the element has a positive or negative impact on flood risk. Equation (2) normalizes the DEM, slope, and distance to river components, whereas equation (1) normalizes the remaining factors.

G. Model for ecological assessment

Following other studies [31], [64], [65], a model for the Ecological Quality Index (EQI) was created by integrating all the indicators using Equation 5:

$$EQI_i = \sum_{j=1}^n w_j N_{j,i} \quad (5)$$

from the government, businesses, and landowners were used to determine the weights of the indicators, quantifying their

relative relevance in defining the EQI. The results are summarised in Table II. According to the EQI score of a pixel or piece of land, which ranged from 0 (worst) to 10 (best), the ecological quality of that pixel or piece of land was classified as exceptional (EQI > 6.5), good (EQI between 5.5 and 6.5), fair (EQI between 4.5 and 5.5), or bad (EQI 4.5).

III. RESULTS

A. Impacts of construction of the KKH on LULC

Figure 2 and Table III show maps of the LULC categories for 2013, 2016, 2019, and 2022. With a kappa index of 0.72 in 2013, the LULC mapping had an accuracy rate of 78%. The

accuracy for 2016 was just over the accuracy of the mapping for 2013 (79%), along with a kappa index of 0.72[66], [67]. Meanwhile, the kappa index was 0.71 in 2019, and the accuracy was 77%. The accuracy was the highest in 2022 (82%), and the kappa score was 0.76, which was also the highest compared to the other years [68]. These calculations showed that the average kappa coefficient was 0.73 and the average overall precision was 79%, which shows a valid classification [64]. It was discovered that the topography and the geographic distribution of LULC types were closely related. Areas with level topography were best for growing crops and building transportation networks. As a result, these areas were mostly made up of land used for farming and construction.

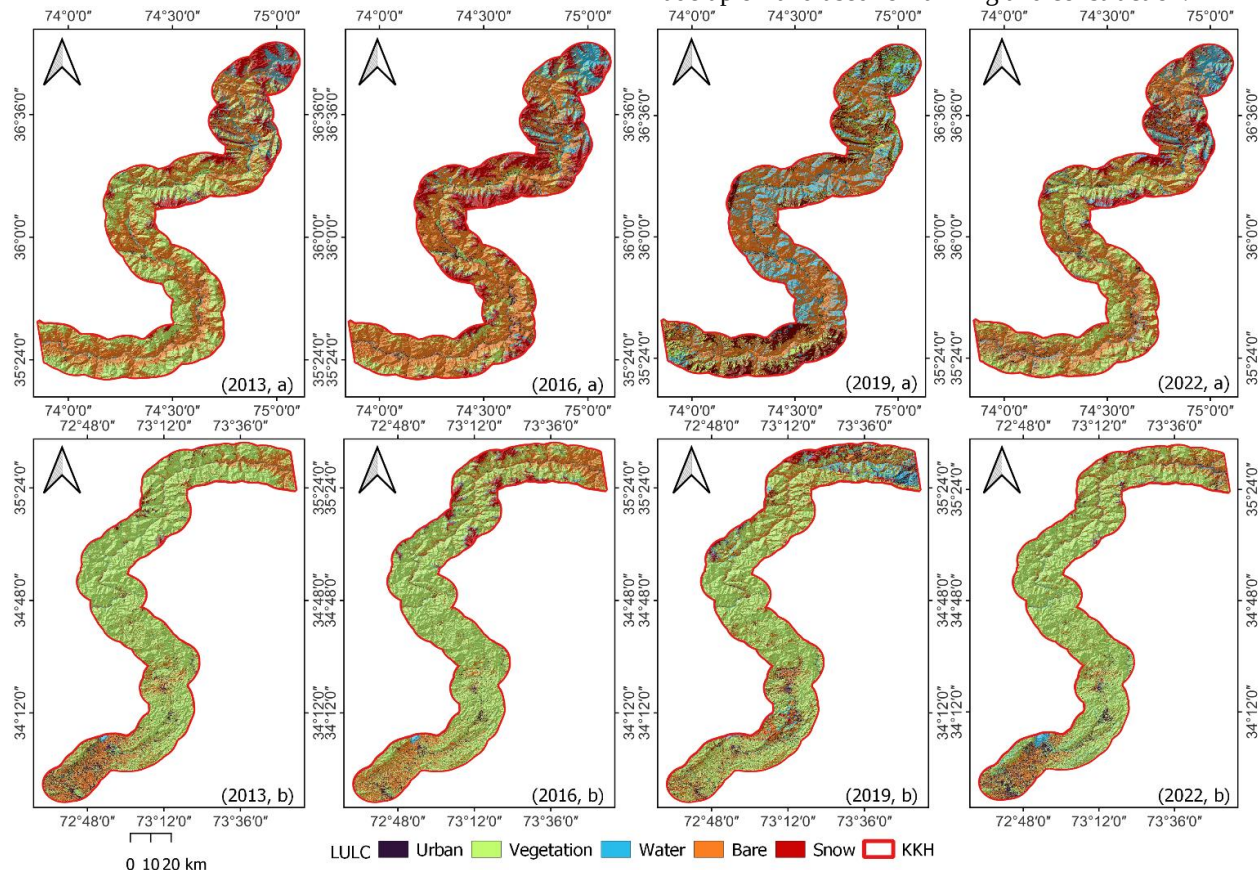


Figure 2. The KKH's changes in LULC between 2013 and 2022

Table III

Accuracy assessment of LULC classification.

Year	Overall Accuracy	Kappa Index
2013	0.78	0.72
2016	0.79	0.72
2019	0.77	0.71
2022	0.82	0.76

Within the used buffer alongside the highway, there was a considerable change in the composition of LULC between 2013 and 2022. Geographically, the compositions of LULC in the buffer showed variable degrees of change intensity during the research. Built-up areas, vegetation, water, bare land, and snow constituted the land cover inside the buffer before the construction of the KKH in 2013. They accounted for 1.7%, 36.31%, 6.93%, 45.67%, and 9.37% for the north portion of

KKH and 4.39%, 73.95%, 1.67%, 19.44%, and 0.53%, for the south portion of KKH correspondingly (Table III). The land cover composition inside the buffer changed significantly after the KKH was finished in 2022, with built-up areas, vegetation, water, bare land, and snow accounting for 4.24%, 29.40%, 11.80%, 46.41%, and 0.75% for the upper portion of KKH, and 5.18%, 70.86%, 3.90%, 19.61% and 0.42% for the lower portion of KKH respectively.

Vegetation coverage suffered the most significant loss out of these five forms of land cover, going from coverage of 36.31% for the upper portion of KKH before the highway's construction to 29.40% after. Similarly, for the lower portion of KKH, the coverage declined from 73.95% to 70.86%. According to the mapping outcomes, the growth of built-up land increased from 1.7% in 2013 to 4.24% in 2022 in the upper portion of the KKH region while increasing from 4.39% in 2013 to 5.18% in 2022 in the lower portion of KKH. This growth represents a net gain of 2.54% for the upper portion of KKH and 0.79% for the lower

portion of KKH. This shows that there was a noticeable change in the built-up land.

According to the figures, forests covered 36.31% of the overall land area in 2013 in the upper portion of KKH, while in the lower portion of KKH, they contributed 73.95% to the overall coverage. However, as seen in Table III, its coverage area fell to 11.60% in 2016 and then increased to 20.81% in 2019 and 29.40% in 2022 for the upper portion of KKH. Similarly, the area for vegetation for the lower portion of KKH fell to 66.17% in 2016 and 62.67% in 2019 but saw an increase to 70.86% in 2022.

TABLE IV
THE CHANGES IN LULC BETWEEN 2013 AND 2022.

Classes	The north portion of KKH				The South portion of KKH			
	2013	2016	2019	2022	2013	2016	2019	2022
Urban	146604 (1.7)	300100 (3.48)	257925 (2.99)	365063 (4.24)	392536 (4.39)	242263 (2.71)	443904 (4.96)	463151 (5.18)
Vegetation	3122800 (36.31)	998166 (11.60)	1789632 (20.81)	2529137 (29.40)	6608139 (73.95)	5913324 (66.17)	5600422 (62.67)	6332401 (70.86)
Water	596053 (6.93)	927651 (10.78)	948719 (11.03)	1015546 (11.80)	149851 (1.67)	230612 (2.58)	533043 (5.96)	349215 (3.90)
Bare	3927856 (45.67)	4647269 (54.04)	4166618 (48.45)	3991571 (46.41)	1737564 (19.44)	2246322 (25.13)	2004960 (22.43)	1752652 (19.61)
Snow	806337 (9.37)	1726464 (20.07)	1436756 (16.70)	698333 (8.12)	47497 (0.53)	303066 (3.39)	353258 (3.95)	38168 (0.42)

The bare land coverage remained comparatively steady compared to the other five land cover categories. From 45.67% in 2013 to 46.41% in 2022, the bare area increased for the northern portion of KKH. Following a similar pattern, the buffer around the lower portion of KKH showed a slight increase from 19.44% in 2013 to 19.61% in 2022. Like bare land, water areas also showed a similar tendency, albeit with a more significant shift of about 4.87% for the north portion of KKH and 2.23% for the south portion of KKH. As per Table

IV, the overall percentage of snow coverage was 4.95% for both the upper portion of KKH and the south portion of KKH in 2013. The area coverage increased from 9.37% in 2013 to 20.07% in 2016 but decreased in 2019 and 2022 to 16.70% and 8.12%, respectively, for the upper portion of KKH. For the lower portion of KKH, the snow coverage increased to 3.39% and 3.95% in 2016 and 2019 but reflected a significant drop to 0.42% in 2022.

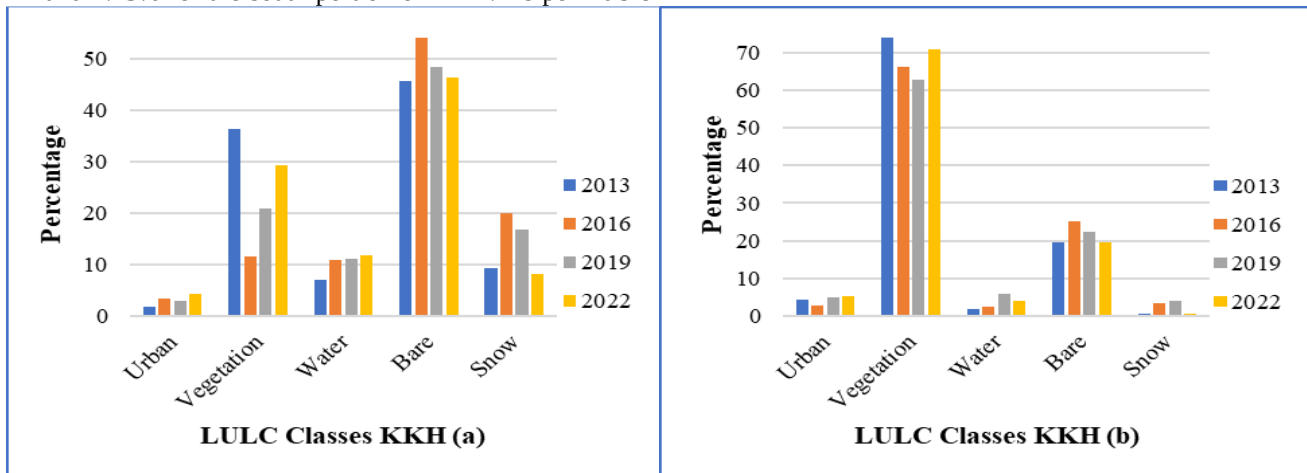


Figure 3. LULC compositions along the KKH from 2013 to 2022.

The LULC in the considered buffer had significant alterations due to the highway's construction. The developed area rose from 5.20% to 14.20% within the buffer, whereas the area covered by forest dropped from 74.60% in 2013 to 69.00% in 2018. The KKH's extension mainly resulted in a 3.50% and

a 5.6% decline in agriculture and woodland, respectively. Like the alterations in built-up, the vegetation land cover also changed the considered time frame, with forest cover predominating from 2013 to 2018. However, a lesser fraction of the overall area comprised built-up land. Figure 3 shows that over the research period, the principal LULC compositions in

the considered buffer remained more consistent apart from vegetation and bare classes, which showed considerable deflections over the years.

B. Topographical settings of the area

The elevation in the northern portion of KKH displays a constant pattern close to the road, staying within the 3000-4000 m range. However, the elevation tends to surge as the distance from the road grows, rising to 4000-6000 m. The elevation even exceeds 6000 m in a few locations on the border of the buffer zone, as shown in Figure 4 (a). Similar patterns can be seen in the elevation for KKH on the south side, which has consistent altitudes between 1000 and 3000 m for areas close to the road.

C. Variations in Ecological Settings

The geographical and temporal fluctuations in LST across the considered zone along the roadway are illustrated in Figure 5. Compared to the start of the considered time of this study, the considered end time showed substantial changes in LST. While the LST in the farthest places from the highway remained unchanging before and after the construction process, the LST in the buffer zone in areas adjacent to the highway grew annually during the same period. The LST in the southern portion of the KKH showed a considerable rise, surpassing 70°C compared to the prior range of 65-70°C, from the start of

However, the elevation steadily rises and reaches a range of 3000 – 4000m as the distance from the road grows. The buffer zone's periphery also has some areas with elevations higher than 4000 m, as seen in Figure 4 (b). The slope degree follows a constant pattern, mainly for elevation across the lower and upper portions of KKH. The slope stays below 15 degrees near the road. The slope degree, however, sharply increases with increasing distance from the road, especially at the margins of the buffer zone, reaching values of more than 60 degrees.

highway construction in 2013 until reaching its peak in 2019. Similarly, in the upper portion of the KKH, the pattern was observed to be the same as LST, showing an increment from 2013 to 2019. Nevertheless, in 2022, a slight decrease in LST was observed on the northern and southern sections of the highway as LST dropped back in the 65-70°C range and, in some small regions, even below the 60°C mark. It is important to note that there was a rise in LST variably during highway construction from 2013 to 2019, which persisted in some regions close to the highway even after the project was finished, as seen in Figure 5.

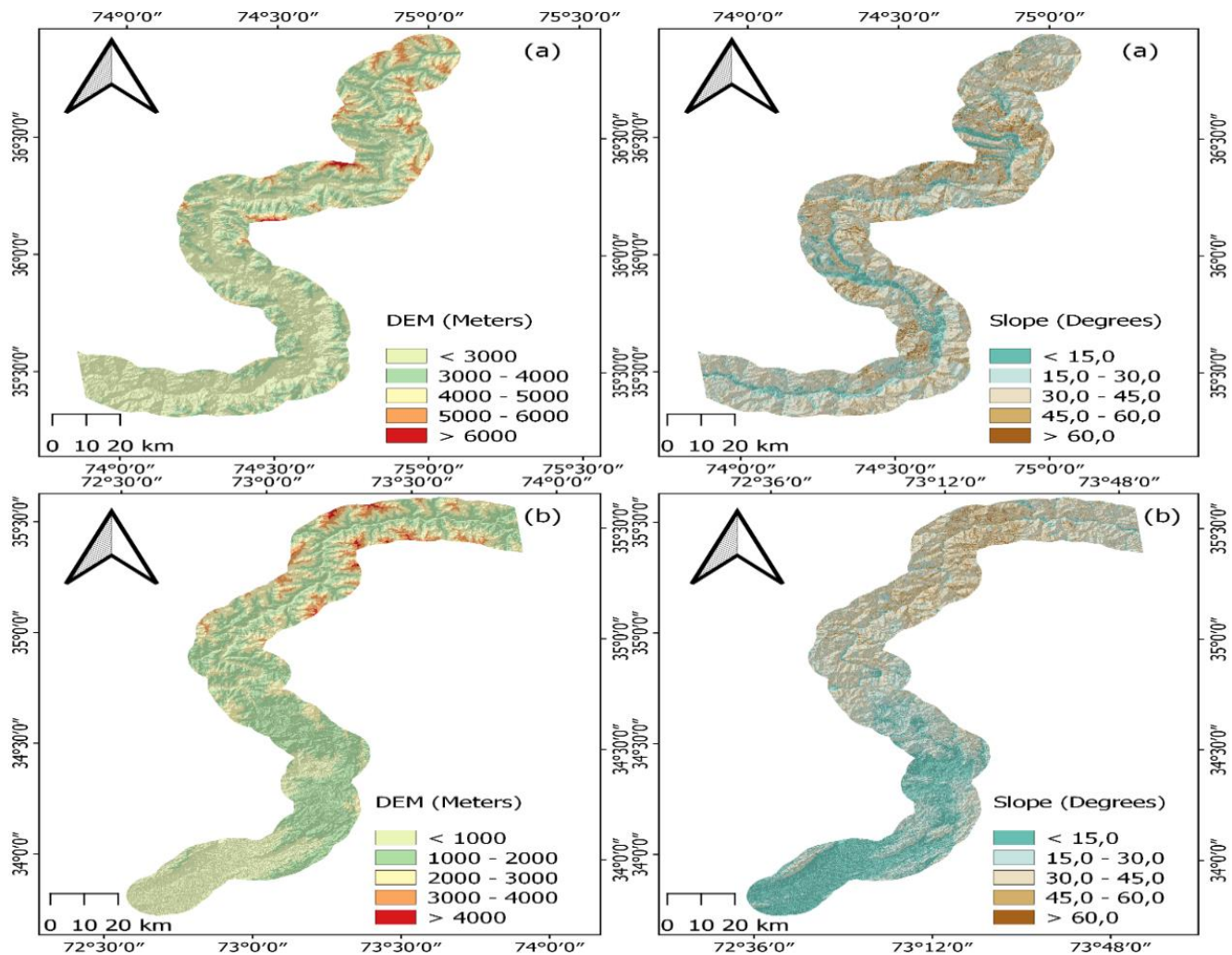


Figure 4. DEM and slope variations along the KKH highway

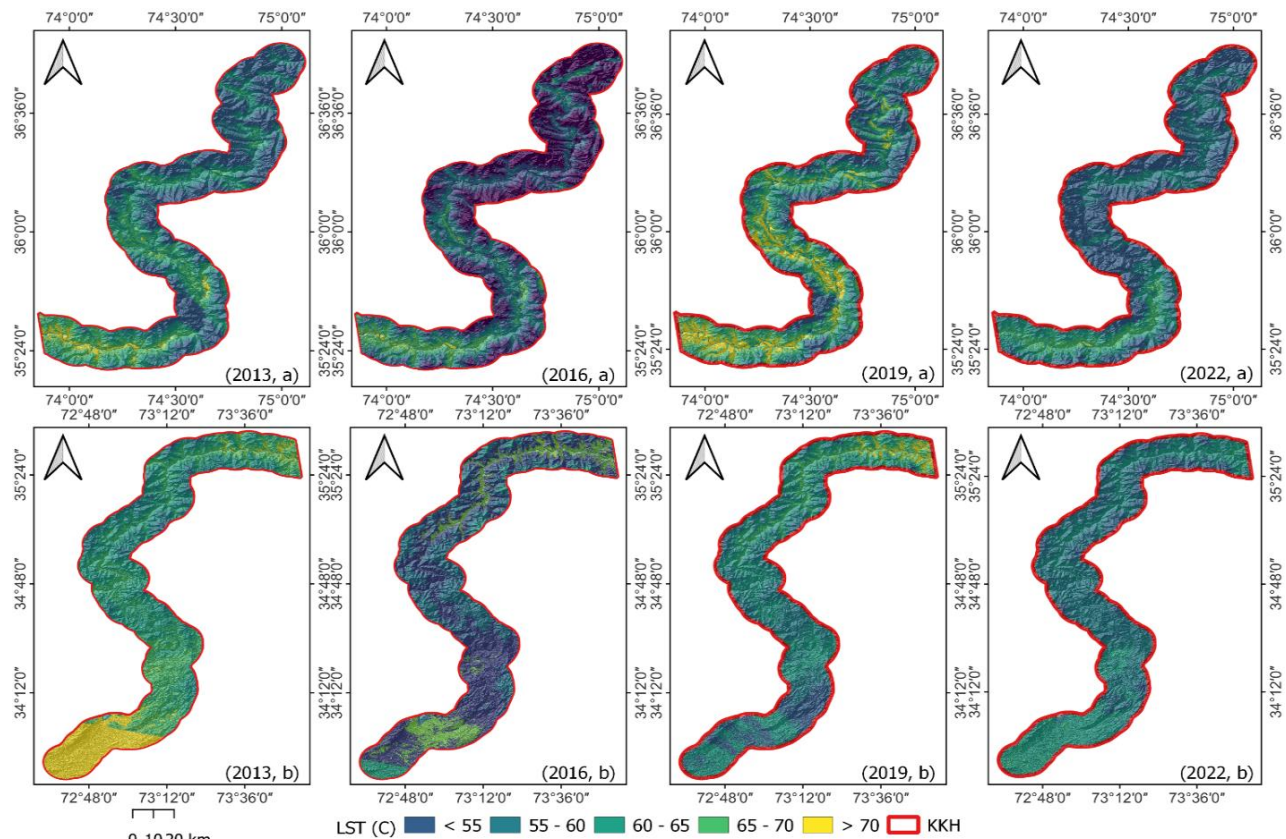


Figure 5. LST variations along the KKH highway from 2013 to 2022.

The trend of NDMI in the buffer zone from 2013 to 2022 exhibits irregular fluctuations, with periods of both increasing and decreasing values, as shown in Figure 6. As the construction of KKH started in 2013 and ended in 2019, the NDMI shows a slight decline overall, especially in the upper part of the KKH buffer. For the upper portion of KKH, the NDMI values consistently fall within the range of 0.1–0.2 near the vicinity of the highway but progressively increase beyond 0.3 as the distance from the highway increases. Regarding the lower portion of KKH, the NDMI values underwent distinct changes over time [69], [70], [71]. In 2013, the values ranged from 0.0 to 0.1, which increased from 0.1 to 0.2 in 2016. However, a decrease was observed in 2019, followed by another increase in 2022. With the NDMI falling from 2013 to 2019 and then progressively rising in 2022, the decline in the NDMI in the buffer zone was considerably less pronounced but still significant. Despite a modest decrease in 2016, the NDMI in the upper portion of the KKH buffer zone mainly remained steady, but the lower portion of KKH showed distinct changes over time. Minimal variation was seen in the buffer zone's NDMI for 2019. Overall, inside the buffer zone, the value of the NDMI considerably rose through the course of highway building.

Figure 7 illustrates the NDVI for the considered buffer zone along the highway for several years. The NDVI values in the upper side of the KKH buffer zone show very constant features from 2013 to 2022. The NDVI readings constantly stay in the 0.0–0.1 range close to the highway, but further away from the road, they rise to the 0.1–0.2 range and, in some places, even surpass 0.3. The NDVI measurements over time show a distinct

trend for the south side of the KKH buffer zone. The NDVI has increased significantly from 2013 to 2016.

On the other hand, a sizable fall was noted in 2019. However, there will be a subsequent increase in 2022, with NDVI values above 0.3 in the more significant part of the buffer zone. The NDVI values in the buffer zone witnessed a minor decline before remaining largely steady. Overall, the variations had significant patterns for the upper and lower portions of the KKH buffer zone. While the trend for the north portion of the KKH was mainly stable, the trend for the south portion of the KKH fluctuated more but had a discernible increase in 2022.

D. Overall variation in EQI

Figures 8, 9, and Table V show that less than 16% of the region had very poor EQI in 2013, rapidly dropping to about 4.31% by 2022. In contrast, the proportion of excellent EQI declined from 2.99% in 2013 to 1% in 2022 for the upper portion of KKH but increased from 0.57% in 2013 to 3.83% in 2022 for the lower portion of KKH. In contrast, the proportion of good EQI dropped from 15.76% in 2013 to 14.15% in 2022 for the upper portion of KKH, whereas for the lower portion of KKH, it rose to 39.76% in 2022 from 9.98% in 2013. The percentage of fair EQI rose from 32.48% in 2013 to 37.87% in 2022 for the upper portion of KKH, and for the lower portion of KKH, it increased from 30.23% to 37.84% for the years 2013 and 2022, respectively. Following an opposite pattern to the one noticed for fair, the percentage of poor decreased from 41.25% in 2013 to 40.94% in 2022 for the upper portion of KKH and from 33.13% in 2012 to 15.99% in 2022 for the lower portion of KKH.

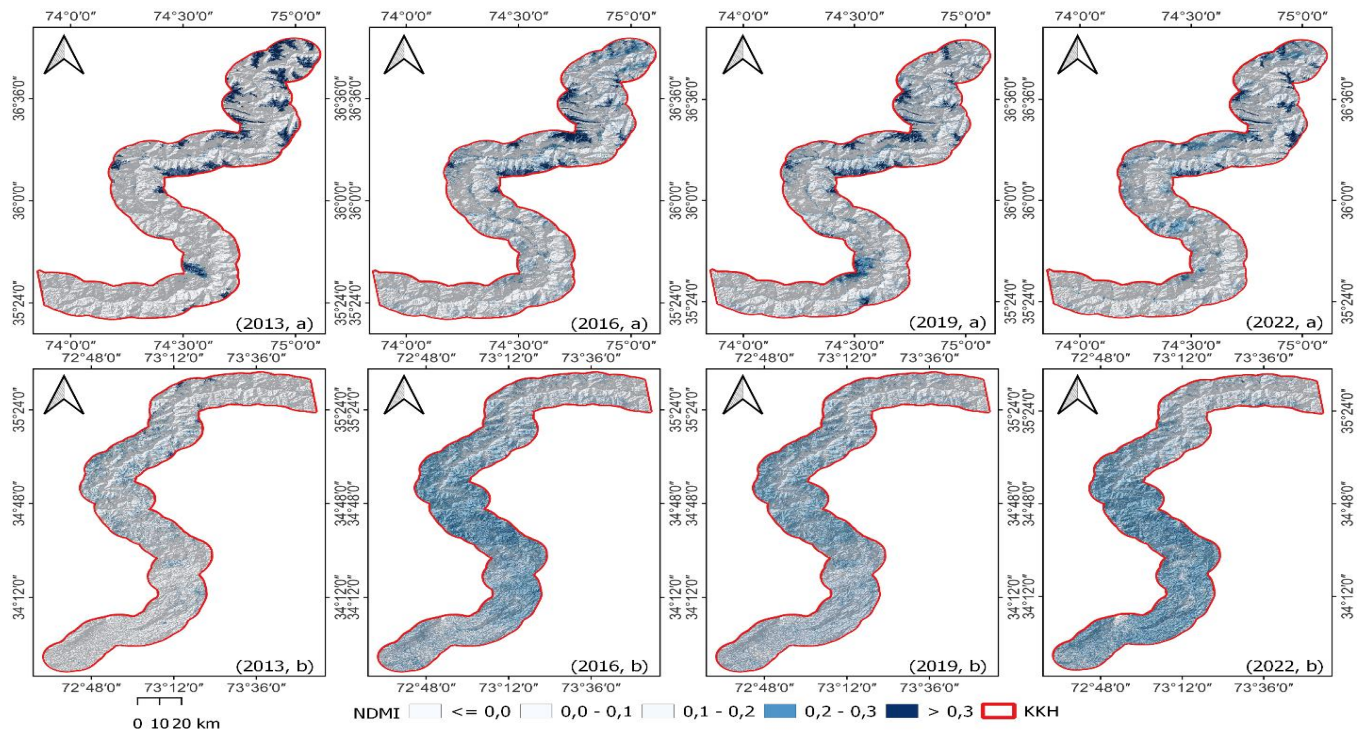


Figure 6. NDMI changes along the KKH from 2013 to 2022.

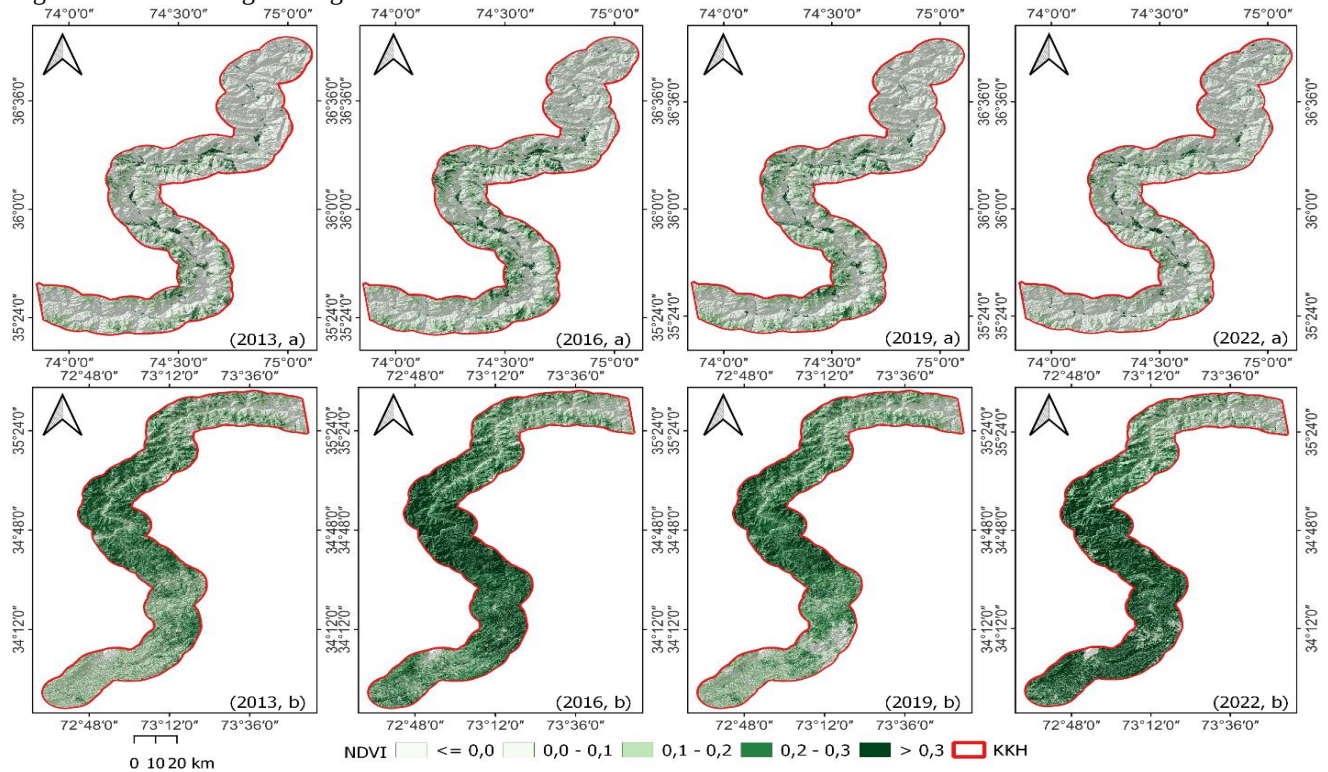


Figure 7. NDVI changes along the KKH from 2013 to 2022.

In contrast to what was seen in the buffer zones of the lower and upper sides of KKH, there were significant changes in the distribution of EQI classes, namely, poor, fair, and excellent, compared to the classes, namely, very poor and excellent. Even though the extent of change was less substantial than those found in other courses, the patterns of change for the very poor and excellent classes were still reasonable. Overall, the poor

EQI share proportions rose with time, from 2013 to 2019, before falling in 2022. Similarly, the fair EQI's overall share rose from 2013 to 2019. An equivalent trend of change might be seen in the good EQI class. The decline and surge in the excellent classes was less than 7% overall. After surging from almost 2% in 2013 to 6% in 2016, the proportion with excellent EQI stayed practically constant for 2019 and 2022.

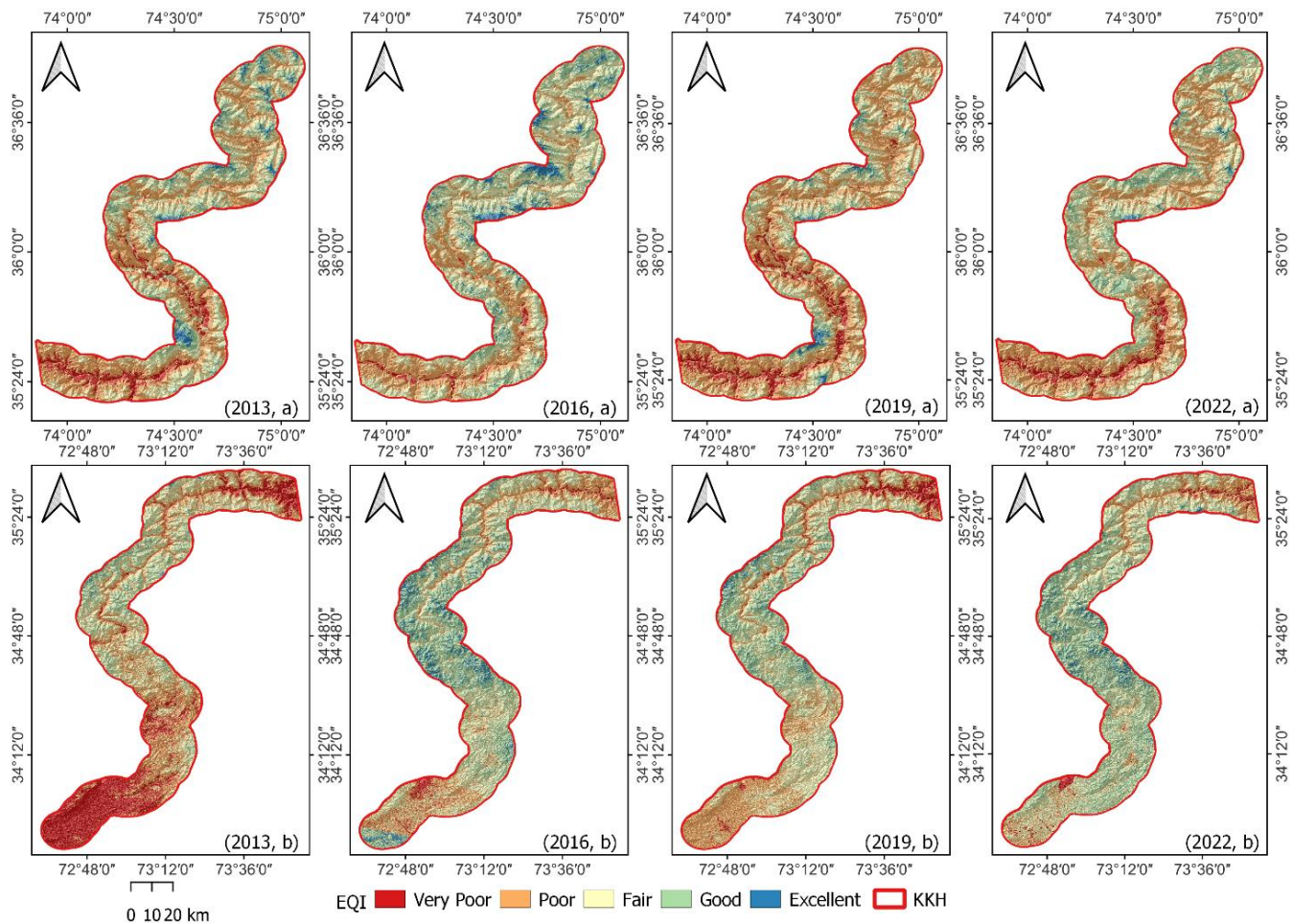


Figure 8. The EQI along KKH from 2013 to 2022.

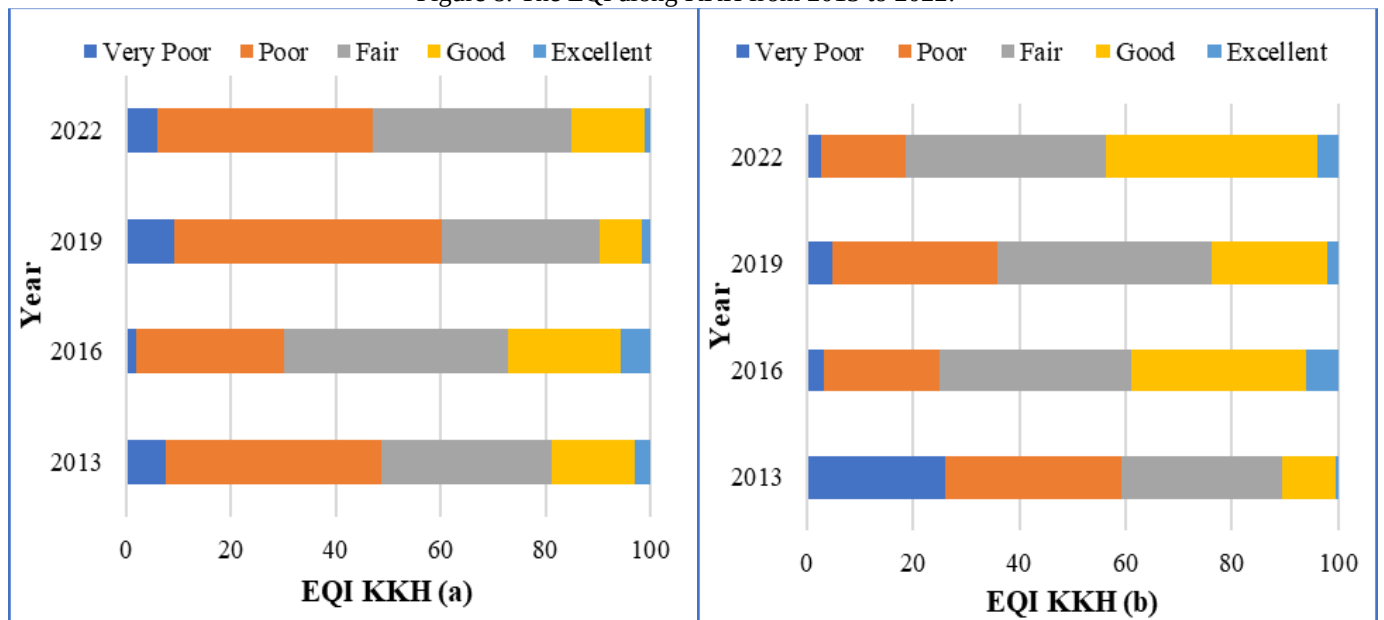


Figure 9. Change in the overall EQI from 2013 to 2022 along KKH.

Except for the poor and excellent categories, which increased and dropped from 2013 to 2022, the changes in the four EQI classes inside the buffer zones for the portions of the KKH were comparable. However, the area for the combined poor and fair EQI classes was significantly present for almost all the years.

The percentage for suitable EQI classes, on the other hand, was the third most prominent class from 2013 to 2018. Overall, the EQI transformation rates were significantly lower throughout the area.

TABLE V
THE ASSESSMENT OF EQI ALONG KKH FROM 2013 TO 2022.

Classes	The upper portion of KKH				The lower portion of KKH			
	2013	2016	2019	2022	2013	2016	2019	2022
Very Poor	580288 (7.52)	133457 (1.94)	703796 (9.14)	465268 (6.04)	1905088 (26.09)	251957 (3.09)	355640 (4.92)	188882 (2.58)
Poor	3180815 (41.25)	1956983 (28.29)	3940351 (51.08)	3158140 (40.94)	2419986 (33.13)	1774794 (21.78)	2253664 (30.85)	1168313 (15.99)
Fair	2504985 (32.48)	2950115 (42.66)	2316216 (30.03)	2921145 (37.87)	2208227 (30.23)	2947364 (36.18)	2952588 (40.42)	2764308 (37.84)
Good	1215761 (15.76)	1483422 (21.45)	623935 (8.08)	1092028 (14.15)	729246 (9.98)	2673475 (32.82)	1596832 (21.86)	2904385 (39.76)
Excellent	229066 (2.99)	391412 (5.66)	128359 (1.67)	75711 (1)	40187 (0.57)	498164 (6.13)	145664 (1.95)	278451 (3.83)

IV. DISCUSSION

A. The effect distance of KKH

How far from the road its impacts stretch is crucial for mitigating and managing road construction effects on the surrounding environment [63]. This study considered ecological indicators to map the impacts of the construction of KKH: NDVI, NDMI, slope, elevation, and LST. The spatial variations in the values of the considered indicators inside the buffers in our analysis align with the highway's construction. The NDVI, NDMI, and LST significantly altered the buffer zone throughout the study period.

As a result, the values revealed that the road effect distance was at least 10 km from the highway. The distance seems a lot because the towns and cities in that region are found primarily alongside the KKH. Thus, the KKH's construction brought in many changes in the area's natural and built-up settings. This result is in line with earlier research that discovered that desert tortoises (*Gopherus agassizii*) affect distances of 400 m [72], for five anuran species, 250–1000 m [63], for great bustards (*Otis tarda*), 1300 m [73], and LULC categories, 400 m [74].

The most surprising thing was that the road-effect distance indicated by variations in spatial patterns of these three indicators could stretch up to 10 km. Moreover, the spatial variations still demonstrated an apparent alteration in the buffer even after 2019. This contrasted with the conclusion drawn from the spatial values, which indicated that there had been no appreciable changes in the range in areas at a distance of 10 km from the KKH. As an illustration, even if the NDVI and NDMI improved inside the 10 km region in 2022, the spatial variability of these two indicators did not. As a result, the findings showed that it was improper to establish the road-effect distance using only the one-time-period circumstances[75], [76], [77].

Our findings revealed that the KKH's effect distance was at least 10 km from the highway itself, as evidenced by the variations in the indicators' values, which demonstrate their spatial and temporal variations. The road-effect zone for the NDVI, NDMI, and LST altered in the buffer zone over time due to the recovery and protection policies implemented in the latter stages[78], [79], [80]. These results align with the findings for the Qinghai-Tibet Plateau [20], research examining highway and railway construction projects affecting vegetation and permafrost degradation settings. The study discovered that the

engineering events had little to no impact on the vegetation outside a region of 10 km or around 15 km alongside the linear highway infrastructure. Additionally, our findings aligned with the road-effect zones for animal movements, which often outspread tens to thousands of meters from highways [3]. In the opposite direction from the road, the combined impacts of highway building on temperature, surface wetness, and vegetation coevolved and coexisted [81], [82].

For instance, as demonstrated in this study, the NDVI and NDMI significantly reduced inside the 10 km buffers, although the LST gradually climbed throughout the study period. The direct loss of green area, comparable to urban zones, may account for a decline in the NDVI and NDMI, as might indirect effects from highway development, including variations in wetness and temperature settings and dust deposition on plants [83], [84].

By obstructing the reflectance qualities of the primary leaves in the areas closest to the roads, dust deposition on leaf surfaces may lower NDVI values and, hence, lower plant photosynthetic activity [83]. The vegetation up to 400 m from the roadway might be impacted by nitrogen from vehicle NOX emissions [85]. Deterioration of the vegetation exposes the surface more, which increases the absorption of solar radiation [86] and ultimately raises the LST. Using a significant portion of incident solar radiation for water evaporation during evapotranspiration, vegetation also plays a vital role in transforming solar energy [87]. As a result, removing vegetation and dropping surface moisture would increase the temperature of the surface.

The road-effect zone for KKH fluctuated longitudinally, was often asymmetrical, and had ambiguous borders, showing the coevolution of several ecological factors as well as uneven terrain and LULC patches on opposing sides of the KKH, which is to earlier results of different studies [88], [89]. For instance, the road-effect zone spread at least 10 km in the KKH's east and west and on the southern side, where the terrain is relatively flat. The vegetation is sparse, whereas, on the north side, where the terrain is mountainous, and the forest cover is dense, the road-effect zone steadied in the range of 5 km from the highway axis. The EQI demonstrated that the most dramatic changes were located near the highway, where significant construction disruption occurred, in terms of the longitudinal variations in the road-effect distance.

B. Environmental recovery following the construction of the KKH

Highway development that harms ecosystems impacts the surroundings and its elements [13], [46]. Typically, the land is divided by highway-building projects, causing habitat fragmentation, habitat loss, and attrition [90], all of which have negative consequences. This investigation's NDVI and NDMI ecological receptors decreased near the highway zone. In the considered buffer zone during 2013, the maximum relative changes in the NDVI, NDMI, and LST values were around 60%, 40%, and 12%, respectively. The composition of the land cover also saw significant alterations in the considered buffer zone following the building of the highway. This resulted from changing impervious surfaces to natural landscapes, which sped up or brought about harmful processes, including soil erosion [91] and vegetation deterioration [92].

Inside the buffer zone, the NDVI and NDMI demonstrated a recovery of about 3% and 12% every year, correspondingly following the completion of the construction process of KKH in 2019. This recovery was likely aided by different environmental restoration measures, including greening and slope protection, implemented since 2016 to stop vegetation degradation and soil erosion. The considerably modest temporal fluctuations in the EQI along some stretches of the KKH show that the natural woods and artificial mixed forests were effectively preserved. The government programs aimed at enhancing the quality of the woods and the ratios of natural and mixed forests in the area [93], [94] may have accounted for this impact. For instance, temperature, vegetation, and surface wetness may all benefit the environment owing to the restoration programs to stop soil erosion and encourage ecological restoration [65].

C. Overall change of EQI

The methodology employed, comprised of compound indicators such as the NDVI, NDMI, LST, and terrain indicators, successfully indicated the combined influence of KKH development, established as the area variations of the five EQI categories. While the extreme categories demonstrated a sharp decline with the road extension, the fair and low categories saw increased trends from 2013 to 2016. This made sense, given how the development of new highways and the ensuing urbanization had altered land cover and fragmented the landscape. After the highway was finished at the end of 2019, the percentages of regions with low and fair EQI declined while those with exceptional and excellent quality rose. Due to environmental restoration initiatives like greening and slope protection implemented to stop vegetation deterioration and soil erosion, the EQI quickly recovered after the conclusion of highway construction.

This study discovered that the EQI changes were often distributed unevenly over the road zone. The cultivated and undeveloped land growth likely led the EQI in the buffer zone to fall more quickly. Compared to the EQI in the buffer zone of the lower part, the EQI in the upper part remained reasonably consistent across the study period. The low-EQI area in the buffer zone rose from 2013 to 2019; therefore, this does not

imply there was no influence on the buffer area far from the KKH.

However, following the construction of the highway in 2019, the total EQI along the road-effect zone started to improve due to the execution of environmental restoration initiatives, such as the Billion Tree Tsunami Program and the Clean Green Pakistan Movement [95]. The improvement in the Ecological Quality Index (EQI) has been gradual because of the decline in Land Surface Temperature (LST) brought on by the improved (NDVI) and (NDMI) as a result of flooding and snow [96].

D. Significance of the proposed approach for monitoring the EQI

When gathering field samples to determine the effects of human infrastructure on the environment, the before-during-after control-impact strategy was frequently utilised [97]. This study's proposed technique can help lessen the consequences of spatial and temporal variability by removing naturally variable temporal impacts. This is a significant benefit since tackling the critical temporal variations in environmental indicators before, during, and after the building of infrastructures is one of the fundamental issues in measuring the environmental consequences of engineering activities [97], [98]. However, because of the requirement for repeated sampling over time, which is costly and challenging, there are relatively few published instances of applying such techniques for evaluating the environmental consequences of roadways [65]. Using classic techniques to assess the impacts of highway construction can be challenging. Still, the approach suggested in this work might complement the classic approaches by including remote sensing capabilities in the inclusive valuation of highways' environmental impacts. The proposed approach might illustrate environmental variations throughout the building and operational process of different infrastructures, which is crucial to assisting decision-making besides mitigation efforts.

Someone might track the spatial changes in ecological processes brought on by diverse human activities and natural disturbances using individual remote-sensing images [99]. Several multidisciplinary studies have been carried out to assess the possible ecological impacts of roads. However, estimating the temporal and spatial variations might be complicated by utilizing rare remotely sensed images, for instance, in humid regions with heavy cloud cover [100]. To reduce the effects of the geographical and temporal variability of the obtained photos, the images were first normalized by utilizing the data from the adjacent reference regions instead of directly using the remote sensing images. By normalizing the remote sensing images, all the images become similar in time and space. The proposed technique in this study is general and may be used to evaluate the effects of any type of disturbance and technical activity, including highway construction.

Following the method proposed in this work, it is possible to quantitatively describe the distances between roads and environmental indicators that may be generated from remote sensing data. Along the highways, the road-effect distances of other crucial indicators, such as air pollution, traffic noise

pollution, animal migration, etc., still rely on field-based measurements. Additionally, the approach's outcomes may be influenced by the standard of the selected reference zone. As a result, efforts should be made to identify the appropriate reference zone that best captures the natural environment and is as close as feasible to the road under investigation.

V. CONCLUSIONS

The current research work implemented a framework to quantify the impacts of the construction of KKH by utilizing remote sensing data. The implemented approach can be easily extended to different places to illustrate and examine the temporal and spatial variations of other environmental impacts caused by highway development. When reviewing the impact distance of the KKH, we discovered that the variation in the spatial inconsistency of ecological circumstances varied concurrently. The spatial alterations of the considered indicators, comprising NDVI, NDMI, LST, and LULC, showed that the KKH's influence distance was at least 10k m from the roadway, as illustrated by the indicators' geographic variations in spatial variability). The construction of the KKH significantly impacted the LULC settings along the KKH in contrast to the other indicators. The considered buffer zones' vegetation and moisture conditions, as demonstrated by NDVI and NDMI, respectively, clearly showed decrease-increase (degradation-recovery) trends that matched well with the development and operation phases of the KKH while the LST gradually increased. According to a compound environment quality score, called the EQI, which considered the slope, elevation, NDVI, LST, and NDMI, the environmental effects of the KKH development were most noticeable when they were close by and diminished with distance.

VI. ACKNOWLEDGEMENT

The authors acknowledge the Researchers Supporting Project number (RSPD2025R951), King Saud University, Riyadh, Saudi Arabia.

VII. REFERENCES

- [1] K. Wu and X.-M. Li, "Deep learning for retrieving omni-directional ocean wave spectra from spaceborne synthetic aperture radar," *Remote Sens Environ*, vol. 314, p. 114386, 2024, doi: <https://doi.org/10.1016/j.rse.2024.114386>.
- [2] G. Cai, X. Zheng, J. Guo, and W. Gao, "Real-time identification of borehole rescue environment situation in underground disaster areas based on multi-source heterogeneous data fusion," *Saf Sci*, vol. 181, p. 106690, 2025, doi: <https://doi.org/10.1016/j.ssci.2024.106690>.
- [3] A. Torres, J. A. G. Jaeger, and J. C. Alonso, "Assessing large-scale wildlife responses to human infrastructure development," *Proc Natl Acad Sci U S A*, vol. 113, no. 30, pp. 8472–8477, Jul. 2016, doi: 10.1073/PNAS.1522488113.
- [4] M. Giunta, "Assessment of the environmental impact of road construction: Modelling and prediction of fine particulate matter emissions," *Build Environ*, vol. 176, p. 106865, 2020.
- [5] Y. QIAO, T. WANG, J. LÜ, and K. LIU, "TEMPO: Time-evolving multi-period observational anomaly detection method for space probes," *Chinese Journal of Aeronautics*, p. 103426, 2025, doi: <https://doi.org/10.1016/j.cja.2025.103426>.
- [6] W. Jing, L. Shao-Qun, T. Dao-Yuan, Y. Jian-Hua, Z. Hong-Hu, and K. S. Chi, "A Novel Method for Integrity Assessment of Soil-Nailing Works with Actively Heated Fiber-Optic Sensors," *Journal of Geotechnical and Geoenvironmental Engineering*, vol. 150, no. 8, p. 4024063, Aug. 2024, doi: 10.1061/JGGEFK.GTENG-11790.
- [7] G. Xiao *et al.*, "PPP ambiguity resolution based on factor graph optimization," *GPS Solutions*, vol. 28, no. 4, p. 178, 2024, doi: 10.1007/s10291-024-01715-6.
- [8] P. L. Ibisch *et al.*, "A global map of roadless areas and their conservation status," *Science (1979)*, vol. 354, no. 6318, pp. 1423–1427, Dec. 2016, doi: 10.1126/SCIENCE.AAF7166.
- [9] X. Ma *et al.*, "A Pixel Dichotomy Coupled Linear Kernel-Driven Model for Estimating Fractional Vegetation Cover in Arid Areas From High-Spatial-Resolution Images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–15, 2023, doi: 10.1109/TGRS.2023.3289093.
- [10] H. Pan, S. Tong, X. Wei, and B. Teng, "Fatigue state recognition system for miners based on a multi-modal feature extraction and fusion framework," *IEEE Trans Cogn Dev Syst*, pp. 1–10, 2024, doi: 10.1109/TCDS.2024.3461713.
- [11] T. Richard, T. Forman, and R. D. Deblinger, "The ecological road-effect zone of a Massachusetts (U.S.A.) suburban highway," *Conservation Biology*, vol. 14, no. 1, pp. 36–46, Feb. 2000, doi: 10.1046/J.1523-1739.2000.99088.X.
- [12] E. Wang *et al.*, "A New Data Completion Perspective on Sparse CrowdSensing: Spatiotemporal Evolutionary Inference Approach," *IEEE Trans Mob Comput*, vol. 24, no. 3, pp. 1357–1371, 2025, doi: 10.1109/TMC.2024.3480983.
- [13] S. Wang, J. Abbas, K. I. Al-Sulati, and S. A. R. Shah, "The impact of economic corridor and tourism on local community's quality of life under one belt one road context," *Eval Rev*, vol. 48, no. 2, pp. 312–345, 2024.
- [14] U. S. A.) Suburban, R. T. T. Forman, and R. D. Deblinger, "Society for Conservation Biology The Ecological Road-Effect Zone of a Massachusetts (," 2000.
- [15] G. Shannon, L. M. Angeloni, G. Wittemyer, K. M. Fristrup, and K. R. Crooks, "Road traffic noise modifies behaviour of a keystone species," *Anim Behav*, vol. 94, pp. 135–141, 2014, doi: 10.1016/J.ANBEHAV.2014.06.004.
- [16] F. Creutzig *et al.*, "Transport: A roadblock to climate change mitigation?," *Science (1979)*, vol. 350, no.

- 6263, pp. 911–912, Nov. 2015, doi: 10.1126/SCIENCE.AAC8033.
- [17] H. Zhang, P. Xiao, F. Yao, Q. Zhang, and Y. Gong, "Fusion of multi-scale attention for aerial images small-target detection model based on PARE-YOLO," *Sci Rep*, vol. 15, no. 1, p. 4753, 2025, doi: 10.1038/s41598-025-88857-w.
- [18] B. Tu, Q. Ren, J. Li, Z. Cao, Y. Chen, and A. Plaza, "NCGLF2: Network combining global and local features for fusion of multisource remote sensing data," *Information Fusion*, vol. 104, p. 102192, 2024, doi: <https://doi.org/10.1016/j.inffus.2023.102192>.
- [19] C. F. Wu, Y. P. Lin, L. C. Chiang, and T. Huang, "Assessing highway's impacts on landscape patterns and ecosystem services: A case study in Puli Township, Taiwan," *Landsc Urban Plan*, vol. 128, pp. 60–71, 2014, doi: 10.1016/j.landurbplan.2014.04.020.
- [20] Y. Song, L. Jin, and H. Wang, "Vegetation changes along the Qinghai-Tibet Plateau Engineering Corridor since 2000 induced by climate change and human activities," *Remote Sens (Basel)*, vol. 10, no. 1, Jan. 2018, doi: 10.3390/RS10010095.
- [21] J. C. Assis, H. C. Giacomini, and M. C. Ribeiro, "Road Permeability Index: Evaluating the heterogeneous permeability of roads for wildlife crossing," *Ecol Indic*, vol. 99, pp. 365–374, Apr. 2019, doi: 10.1016/J.ECOLIND.2018.12.012.
- [22] B. Zhang *et al.*, "A Multi-Baseline Forest Height Estimation Method Combining Analytic and Geometric Expression of the RVoG Model," 2024, doi: 10.3390/f15091496.
- [23] S. Gulci, A. Emin, A. Bursa, T. Üniversitesi, and H. Oguz, "ASSESSMENT OF THE ROAD IMPACTS ON CONIFEROUS SPECIES WITHIN THE ROAD-EFFECT ZONE USING NDVI ANALYSIS APPROACH Investigating Negative Effects of Mechanized Harvesting Machines on Forest Soil and Developing Methods to Minimize These Effects View project Developing GIS database for Baskonus Research and Application Forest of KSU View project." [Online]. Available: <https://www.researchgate.net/publication/313512167>
- [24] H. Alphan and M. A. Derse, "Change detection in Southern Turkey using normalized difference vegetation index (NDVI)," *Journal of Environmental Engineering and Landscape Management*, vol. 21, no. 1, pp. 12–18, Mar. 2013, doi: 10.3846/16486897.2012.663091.
- [25] N. Zi, X.-M. Li, M. Gade, H. Fu, and S. Min, "Ocean eddy detection based on YOLO deep learning algorithm by synthetic aperture radar data," *Remote Sens Environ*, vol. 307, p. 114139, 2024, doi: <https://doi.org/10.1016/j.rse.2024.114139>.
- [26] Z. Chen *et al.*, "Remote sensing research on plastics in marine and inland water: Development, opportunities and challenge," *J Environ Manage*, vol. 373, p. 123815, 2025, doi: <https://doi.org/10.1016/j.jenvman.2024.123815>.
- [27] B. Tariq, "Environmental Impacts of CPEC Roadway Network and Mitigation Techniques."
- [28] S. Feng *et al.*, "Quantification of the environmental impacts of highway construction using remote sensing approach," *Remote Sens (Basel)*, vol. 13, no. 7, Apr. 2021, doi: 10.3390/rs13071340.
- [29] Y. Zhao, X. Wang, and Z. Huang, "Multi-Function Radar Modeling: A Review," *IEEE Sens J*, vol. 24, no. 20, pp. 31658–31680, 2024, doi: 10.1109/JSEN.2024.3436877.
- [30] L. I. Khan, "Environmental Impact Assessment of Faisalabad-Khanewal Motorway (M-4)," 2012.
- [31] K. Soanes *et al.*, "Do wildlife crossing structures mitigate the barrier effect of roads on animal movement? A global assessment," *Journal of Applied Ecology*, vol. 61, no. 3, pp. 417–430, 2024.
- [32] H. Gucinski, *Forest roads: a synthesis of scientific information*, vol. 509. US Department of Agriculture, Forest Service, Pacific Northwest Research Station, 2001.
- [33] L. Wang *et al.*, "Research on high precision localization of space target with multi-sensor association," *Opt Lasers Eng*, vol. 184, p. 108553, 2025, doi: <https://doi.org/10.1016/j.optlaseng.2024.108553>.
- [34] V. Nedbal and J. Brom, "Impact of highway construction on land surface energy balance and local climate derived from LANDSAT satellite data," *Science of the total environment*, vol. 633, pp. 658–667, 2018.
- [35] C. Jiang, Y. Wang, Z. Yang, and Y. Zhao, "Do adaptive policy adjustments deliver ecosystem-agriculture-economy co-benefits in land degradation neutrality efforts? Evidence from southeast coast of China," *Environ Monit Assess*, vol. 195, no. 10, p. 1215, 2023, doi: 10.1007/s10661-023-11821-6.
- [36] J. Liu *et al.*, "Soil ecological stoichiometry synchronously regulates stream nitrogen and phosphorus concentrations and ratios," *Catena (Amst)*, vol. 231, p. 107357, 2023, doi: <https://doi.org/10.1016/j.catena.2023.107357>.
- [37] H. Xu, Q. Li, and J. Chen, "Highlight Removal from A Single Grayscale Image Using Attentive GAN," *Applied Artificial Intelligence*, vol. 36, no. 1, p. 1988441, Dec. 2022, doi: 10.1080/08839514.2021.1988441.
- [38] G. Zhou and X. Liu, "Orthorectification Model for Extra-Length Linear Array Imagery," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–10, 2022, doi: 10.1109/TGRS.2022.3223911.
- [39] G. Zhou, W. Liu, Q. Zhu, Y. Lu, and Y. Liu, "ECA-MobileNetV3(Large)+SegNet Model for Binary Sugarcane Classification of Remotely Sensed Images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–15, 2022, doi: 10.1109/TGRS.2022.3215802.
- [40] D. Xie *et al.*, "Aboveground Biomass Prediction of Arid Shrub-Dominated Community Based on Airborne LiDAR through Parametric and Nonparametric Methods," *Remote Sens (Basel)*, vol.

- 15, no. 13, p. 3344, Jun. 2023, doi: 10.3390/rs15133344.
- [41] X. Chen *et al.*, "Compatible Biomass Model with Measurement Error Using Airborne LiDAR Data," *Remote Sens (Basel)*, vol. 15, no. 14, p. 3546, Jul. 2023, doi: 10.3390/rs15143546.
- [42] S. G. Perz, "The promise and perils of roads," *Nature* 2014 513:7517, vol. 513, no. 7517, pp. 178–179, Aug. 2014, doi: 10.1038/nature13744.
- [43] X. Xu, X. Fu, H. Zhao, M. Liu, A. Xu, and Y. Ma, "Three-Dimensional Reconstruction and Geometric Morphology Analysis of Lunar Small Craters within the Patrol Range of the Yutu-2 Rover," 2023. doi: 10.3390/rs15174251.
- [44] A. Maqsoom *et al.*, "Landslide susceptibility mapping along the China Pakistan Economic Corridor (CPEC) route using multi-criteria decision-making method," *Modeling Earth Systems and Environment* 2021 8:2, vol. 8, no. 2, pp. 1519–1533, Jul. 2021, doi: 10.1007/S40808-021-01226-0.
- [45] M. Ding *et al.*, "Surge-type glaciers in Karakoram Mountain and possible catastrophes alongside a portion of the Karakoram Highway," *Natural Hazards* 2017 90:2, vol. 90, no. 2, pp. 1017–1020, Oct. 2017, doi: 10.1007/S11069-017-3063-4.
- [46] V. Nedbal and J. Brom, "Impact of highway construction on land surface energy balance and local climate derived from LANDSAT satellite data," *Science of the Total Environment*, vol. 633, pp. 658–667, Aug. 2018, doi: 10.1016/j.scitotenv.2018.03.220.
- [47] Y. Chi, Z. Zhang, J. Gao, Z. Xie, M. Zhao, and E. Wang, "Evaluating landscape ecological sensitivity of an estuarine island based on landscape pattern across temporal and spatial scales," *Ecol Indic*, vol. 101, pp. 221–237, Jun. 2019, doi: 10.1016/J.ECOLIND.2019.01.012.
- [48] M. Liu *et al.*, "An Improved Flexible Spatiotemporal DATA Fusion (IFSDAF) method for producing high spatiotemporal resolution normalized difference vegetation index time series," *Remote Sens Environ*, vol. 227, pp. 74–89, Jun. 2019, doi: 10.1016/J.RSE.2019.03.012.
- [49] J. Zhang, L. Zhang, C. Xu, W. Liu, Y. Qi, and X. Wo, "Vegetation variation of mid-subtropical forest based on MODIS NDVI data — A case study of Jinggangshan City, Jiangxi Province," *Acta Ecologica Sinica*, vol. 34, no. 1, pp. 7–12, Feb. 2014, doi: 10.1016/J.CHNAES.2013.09.005.
- [50] S. K. Karan, S. R. Samadder, and S. K. Maiti, "Assessment of the capability of remote sensing and GIS techniques for monitoring reclamation success in coal mine degraded lands," *J Environ Manage*, vol. 182, pp. 272–283, Nov. 2016, doi: 10.1016/J.JENVMAN.2016.07.070.
- [51] H. Zhang, S. M. Gorelick, P. V. Zimba, and X. Zhang, "A remote sensing method for estimating regional reservoir area and evaporative loss," *J Hydrol (Amst)*, vol. 555, pp. 213–227, Dec. 2017, doi: 10.1016/J.JHYDROL.2017.10.007.
- [52] O. Rozenstein, Z. Qin, Y. Derimian, and A. Karnieli, "Derivation of land surface temperature for landsat-8 TIRS using a split window algorithm," *Sensors (Switzerland)*, vol. 14, no. 4, pp. 5768–5780, Mar. 2014, doi: 10.3390/S140405768.
- [53] C. Xiong *et al.*, "Using satellite imagery to estimate CO2 partial pressure and exchange with the atmosphere in the Songhua River," *J Hydrol (Amst)*, vol. 634, p. 131074, 2024, doi: <https://doi.org/10.1016/j.jhydrol.2024.131074>.
- [54] T. W. Owen, T. N. Carlson, and R. R. Gillies, "An assessment of satellite remotely-sensed land cover parameters in quantitatively describing the climatic effect of urbanization," <http://dx.doi.org/10.1080/014311698215171>, vol. 19, no. 9, pp. 1663–1681, Jan. 2010, doi: 10.1080/014311698215171.
- [55] X. L. Chen, H. M. Zhao, P. X. Li, and Z. Y. Yin, "Remote sensing image-based analysis of the relationship between urban heat island and land use/cover changes," *Remote Sens Environ*, vol. 104, no. 2, pp. 133–146, Sep. 2006, doi: 10.1016/J.RSE.2005.11.016.
- [56] W. Li, Q. Cao, K. Lang, and J. Wu, "Linking potential heat source and sink to urban heat island: Heterogeneous effects of landscape pattern on land surface temperature," *Science of The Total Environment*, vol. 586, pp. 457–465, May 2017, doi: 10.1016/J.SCITOTENV.2017.01.191.
- [57] Y. C. Wang, B. K. H. Hu, S. W. Myint, C. C. Feng, W. T. L. Chow, and P. F. Passy, "Patterns of land change and their potential impacts on land surface temperature change in Yangon, Myanmar," *Science of The Total Environment*, vol. 643, pp. 738–750, Dec. 2018, doi: 10.1016/J.SCITOTENV.2018.06.209.
- [58] M. Simwanda, M. Ranagalage, R. C. Estoque, and Y. Murayama, "Spatial Analysis of Surface Urban Heat Islands in Four Rapidly Growing African Cities," *Remote Sensing* 2019, Vol. 11, Page 1645, vol. 11, no. 14, p. 1645, Jul. 2019, doi: 10.3390/RS11141645.
- [59] C. Deng and C. Wu, "Examining the impacts of urban biophysical compositions on surface urban heat island: A spectral unmixing and thermal mixing approach," *Remote Sens Environ*, vol. 131, pp. 262–274, Apr. 2013, doi: 10.1016/J.RSE.2012.12.020.
- [60] W. Liu, J. Zhan, F. Zhao, H. Yan, F. Zhang, and X. Wei, "Impacts of urbanization-induced land-use changes on ecosystem services: A case study of the Pearl River Delta Metropolitan Region, China," *Ecol Indic*, vol. 98, pp. 228–238, Mar. 2019, doi: 10.1016/J.ECOLIND.2018.10.054.
- [61] S. A. Boadi, K. Owusu, and P. B. Obour, "Impact of cashew cultivation on food security, sovereignty, and income diversification in Jaman North District, Ghana," *GeoJournal*, vol. 90, no. 1, p. 36, 2025, doi: 10.1007/s10708-025-11283-y.
- [62] F. Haq, T. Shutkin, M. Afreen, and B. G. Mark, "Cryo-social dynamics: the interplay of glacial dynamics and socioeconomic conditions in the Shigar

- Valley, Karakoram, Pakistan,” *GeoJournal*, vol. 90, no. 1, p. 37, 2025, doi: 10.1007/s10708-025-11289-6.
- [63] F. Eigenbrod, S. J. Hecnar, and L. Fahrig, “Quantifying the road-effect zone: Threshold effects of a motorway on anuran populations in Ontario, Canada,” *Ecology and Society*, vol. 14, no. 1, 2009, doi: 10.5751/ES-02691-140124.
- [64] G. M. Foody, “Explaining the unsuitability of the kappa coefficient in the assessment and comparison of the accuracy of thematic maps obtained by image classification,” *Remote Sens Environ*, vol. 239, p. 111630, Mar. 2020, doi: 10.1016/J.RSE.2019.111630.
- [65] S. Zhang, N. Liu, B. Ma, and S. Yan, “The effects of street environment features on road running: An analysis using crowdsourced fitness tracker data and machine learning,” *Environ Plan B Urban Anal City Sci*, vol. 51, no. 2, pp. 529–545, 2024.
- [66] A. Dutta, M. Banerjee, and R. Ray, “Socio-economic backwardness assessment modelling in different gram panchayats of Sali watershed using multi-criteria based weighting and ranking methods,” *GeoJournal*, vol. 90, no. 1, p. 28, 2025, doi: 10.1007/s10708-024-11276-3.
- [67] S. Liu, “Evaluation of urban land use efficiency in southern Hebei based on data envelopment analysis,” *GeoJournal*, vol. 90, no. 1, p. 40, 2025, doi: 10.1007/s10708-025-11285-w.
- [68] G. Zhou *et al.*, “Off-Axis Four-Reflection Optical Structure for Lightweight Single-Band Bathymetric LiDAR,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–17, 2023, doi: 10.1109/TGRS.2023.3298531.
- [69] M. R. Parvathy and I. K. Manonmani, “Assessing Urban Liveability: A Case Study of Madurai City Using the Indian Liveability Index,” *GeoJournal*, vol. 90, no. 1, p. 46, 2025, doi: 10.1007/s10708-025-11286-9.
- [70] G. M. Kumssa, “The nexus between poverty and environmental degradation in Girar Jarso district, north-central Ethiopia,” *GeoJournal*, vol. 90, no. 1, p. 47, 2025, doi: 10.1007/s10708-025-11294-9.
- [71] F. Sultana, S. A. Akter, Z. Ahmed, and A. Kadir, “Are land use conflicts a barrier to sustainable city development? Evidence from Chattogram District of Bangladesh,” *GeoJournal*, vol. 90, no. 1, p. 45, 2025, doi: 10.1007/s10708-025-11295-8.
- [72] W. I. Boarman and M. Sazaki, “A highway’s road-effect zone for desert tortoises (*Gopherus agassizii*),” *J Arid Environ*, vol. 65, no. 1, pp. 94–101, Apr. 2006, doi: 10.1016/J.JARIDENV.2005.06.020.
- [73] R. T. T. Forman, “Road ecology: a solution for the giant embracing us,” *Landsc Ecol*, vol. 13, no. 4, pp. III–V, 1998.
- [74] C. Wu, Y. Lin, L. Chiang, T. H.-L. and U. Planning, and undefined 2014, “Assessing highway’s impacts on landscape patterns and ecosystem services: A case study in Puli Township, Taiwan,” *Elsevier*, vol. 128, pp. 60–71, 2014, doi: 10.1016/j.landurbplan.2014.04.020.
- [75] Z. Zafar, “Assessment of urbanization impacts on vegetation cover in major cities of Pakistan: evidence from remotely sensed data,” *GeoJournal*, vol. 89, no. 4, p. 167, 2024, doi: 10.1007/s10708-024-11189-1.
- [76] M. Ben-Said, A. Chemchaoui, I. Etebaai, and M. Taher, “Land use and land cover changes in Morocco: trends, research gaps, and perspectives,” *GeoJournal*, vol. 90, no. 1, p. 44, 2025, doi: 10.1007/s10708-024-11214-3.
- [77] O. Hasrama, E. Çaro, and S. Sinjari, “From dominance to autonomy: core-periphery dynamics in Tirana’s municipality,” *GeoJournal*, vol. 90, no. 1, p. 27, 2025, doi: 10.1007/s10708-024-11275-4.
- [78] R. W. Aslam, I. Naz, A. Quddoos, and M. R. Quddusi, “Assessing climatic impacts on land use and land cover dynamics in Peshawar, Khyber Pakhtunkhwa, Pakistan: a remote sensing and GIS approach,” *GeoJournal*, vol. 89, no. 5, p. 202, 2024, doi: 10.1007/s10708-024-11203-6.
- [79] M. Eshghizadeh and M. Mohammadpoor, “An intelligent method to identify land use changes in an urban watershed in Iran,” *GeoJournal*, vol. 90, no. 1, p. 32, 2025, doi: 10.1007/s10708-024-11254-9.
- [80] F. Nahas, “High-Resolution Insights into Nighttime Urban Heat Island Detection: a comparative temporal analysis of 1988 and 2015 in Greater Cairo,” *GeoJournal*, vol. 90, no. 1, p. 30, 2025, doi: 10.1007/s10708-024-11280-7.
- [81] M. Giunta, “Assessment of the environmental impact of road construction: Modelling and prediction of fine particulate matter emissions,” *Build Environ*, vol. 176, Jun. 2020, doi: 10.1016/j.buildenv.2020.106865.
- [82] T. Zhou, X. Luo, Y. Hou, Y. Xiang, and S. Peng, “Quantifying the effects of road width on roadside vegetation and soil conditions in forests,” *Landsc Ecol*, vol. 35, no. 1, pp. 69–81, Jan. 2020, doi: 10.1007/S10980-019-00930-8/FIGURES/10.
- [83] D. E. Ackerman and J. C. Finlay, “Road dust biases NDVI and alters edaphic properties in Alaskan arctic tundra,” *Sci Rep*, vol. 9, no. 1, Dec. 2019, doi: 10.1038/S41598-018-36804-3.
- [84] J. Rahul and M. K. Jain, “An Investigation in to the Impact of Particulate Matter on Vegetation along the National Highway: A Review,” *Res J Environ Sci*, vol. 8, no. 7, pp. 356–372, Jul. 2014, doi: 10.3923/RJES.2014.356.372.
- [85] K. Redling, E. Elliott, D. Bain, and J. Sherwell, “Highway contributions to reactive nitrogen deposition: Tracing the fate of vehicular NO_x using stable isotopes and plant biomonitors,” *Biogeochemistry*, vol. 116, no. 1–3, pp. 261–274, Dec. 2013, doi: 10.1007/S10533-013-9857-X.
- [86] D. Li *et al.*, “Urban heat island: Aerodynamics or imperviousness?,” *Sci Adv*, vol. 5, no. 4, 2019, doi: 10.1126/SCIADV.AAU4299.
- [87] L. Maria Teophilo Aparecido *et al.*, “Higher tree transpiration due to road-associated edge effects in a tropical moist lowland forest,” *Elsevier*, vol. 213, pp. 183–192, Nov. 2015, doi: 10.1016/j.agrformet.2015.06.009.

- [88] J. Liang, Y. Liu, L. Ying, P. Li, Y. Xu, and Z. Shen, "Road impacts on spatial patterns of land use and landscape fragmentation in three parallel rivers region, Yunnan Province, China," *Chin Geogr Sci*, vol. 24, no. 1, pp. 15–27, Feb. 2014, doi: 10.1007/S11769-014-0652-Y.
- [89] S. Pariente, Z. Helena, S. Eyal, F. G. Anatoly, and Z. Michal, "Road side effect on lead content in sandy soil," *Catena (Amst)*, vol. 174, pp. 301–307, Mar. 2019, doi: 10.1016/J.CATENA.2018.11.014.
- [90] R. T. T. Forman and L. E. Alexander, "Roads and their major ecological effects," *Annu Rev Ecol Syst*, vol. 29, pp. 207–231, 1998, doi: 10.1146/ANNUREV.ECOLSYS.29.1.207.
- [91] F. Kleinschroth, N. Laporte, W. F. Laurance, S. J. Goetz, and J. Ghazoul, "Road expansion and persistence in forests of the Congo Basin," *Nat Sustain*, vol. 2, no. 7, pp. 628–634, Jul. 2019, doi: 10.1038/S41893-019-0310-6.
- [92] A. Deljouei, S. Sadeghi, ... E. A.-E. J. of, and undefined 2018, "The impact of road disturbance on vegetation and soil properties in a beech stand, Hyrcanian forest," *Springer*, Accessed: May 11, 2023. [Online]. Available: <https://link.springer.com/article/10.1007/s10342-018-1138-8>
- [93] S. Khalil, "POLICY RECOMMENDATIONS FOR THE IMPACT OF CPEC ON CLIMATE CHANGE; A CASE STUDY OF GILGIT BALTISTAN," 2021.
- [94] U. Ashraf, "Participation and Exclusion in Mega Tree-planting Projects: A Case Study of the Ten Billion Tree Tsunami Programme, Pakistan," 2022, Accessed: May 20, 2023. [Online]. Available: <https://opendocs.ids.ac.uk/opendocs/handle/20.500.12413/17511>
- [95] A. Kamal, A. Ali, M. Yingjie, and A. A. Significance, "of Billion Tree Tsunami Afforestation Project and Legal Developments in Forest Sector of Pakistan," *International Journal of Law and Society*, vol. 1, no. 4, pp. 157–165, 2018, doi: 10.11648/j.ijls.20180104.13.
- [96] R. Ullah, J. Khan, I. Ullah, F. Khan, and Y. Lee, "Assessing Impacts of Flood and Drought over the Punjab Region of Pakistan Using Multi-Satellite Data Products," *Remote Sensing 2023, Vol. 15, Page 1484*, vol. 15, no. 6, p. 1484, Mar. 2023, doi: 10.3390/RS15061484.
- [97] I. Roedenbeck, L. Fahrig, C. Findlay, J. H.-E. and society, and undefined 2007, "The Rauischholzhausen agenda for road ecology," *JSTOR*, Accessed: May 11, 2023. [Online]. Available: <https://www.jstor.org/stable/26267845>
- [98] X. Hu and H. Xu, "A new remote sensing index for assessing the spatial heterogeneity in urban ecological quality: A case from Fuzhou City, China," *Ecol Indic*, vol. 89, pp. 11–21, Jun. 2018, doi: 10.1016/J.ECOLIND.2018.02.006.
- [99] L. Reymondin, K. Argote, A. Jarvis, C. N.-Washington, D. Inter, and undefined 2013, "Road impact assessment using remote sensing methodology for monitoring land-use change in Latin America: Results of five case studies," *terra-i.org*, Accessed: May 11, 2023. [Online]. Available: http://terra-i.org/magnoliaAuthor/dam/jcr:79b0b086-4d5c-4e84-b396-8541e1cec409/Road_Impact_Assessment_Using_Remote_Sensing.pdf
- [100] S. Feng, S. Liu, Z. Huang, ... L. J.-P. of the, and undefined 2019, "Inland water bodies in China: Features discovered in the long-term satellite data," *National Acad Sciences*, Accessed: May 11, 2023. [Online]. Available: <https://www.pnas.org/doi/abs/10.1073/pnas.1910872116>



design.

Bin Zhu received his master's degree from Sichuan Agricultural University in 2007 and graduated from the School of Urban Design, Wuhan University, in 2021. He is currently a professor at the School of Fine Arts and Design at Leshan Normal University. His research covers a wide range of fields, including landscape ecology, environmental ecology, landscape materials, and interior



Ahsen Maqsoom is working as an Associate Professor at the University Mohammed VI Polytechnic (UM6P), located in Benguerir, Morocco. He holds a Ph.D. in Construction Engineering Management and has a strong academic and research background focused on the intersection of renewable energy technologies, economic growth, and sustainability in construction. Dr. Maqsoom's research interests centre around sustainable construction, energy transition strategies, RS and GIS and green technological innovation, with a particular focus on Asian countries. His work has been published in various international journals and conferences. Dr. Maqsoom is actively involved in policy advisory roles, offering expertise to governments and international organizations.



Dr. Lapyote Prasittisopin is an Associate Professor in the Department of Architecture and Director of the Center of Excellence on Green Tech in Architecture at Chulalongkorn University in Bangkok, Thailand. He holds 14 patents and has published more than 80 articles on construction, materials, and sustainability.



Danyal Aslam graduated with a Degree in civil engineering from COMSATS University Islamabad Wah Campus, specializing in transportation and highways. He graduated with distinction and was awarded a gold medal. He has 2 years of experience as a Road Design Engineer. His research interests include sustainable infrastructure, highway engineering, transportation planning,

Remote Sensing, and Road Geometric Design.



Umer Khalil is an enthusiastic and adaptive professional passionate about enhancing his skills and contributing to innovative projects. With a strong foundation in remote sensing, geoinformatics (GIS), and civil engineering, Umer specialises in urban systems, geospatial mapping, and environmental sustainability. Umer is currently working as a GIS engineer in a tech company. His

expertise spans a range of interdisciplinary fields, including urban and regional planning, smart city development, geospatial data analysis, machine learning applications, hazard and risk assessment, and addressing wicked problems in environmental and urban contexts.



Niamat Ullah received the M.Phil. degree in geospatial sciences (GIS & RS) from the National Center of Excellence in Geology (NCEG), the University of Peshawar, Pakistan in 2021. He is currently with the Spatial Decision Support System Lab (SDSSL), National Center of GIS and Space Applications (NCGSA), Pakistan. He is the author of five books, five journal articles, and book chapters. His research interests include

GIS, RS, natural hazards (floods, droughts), hydrology, climate change, and physical geography.



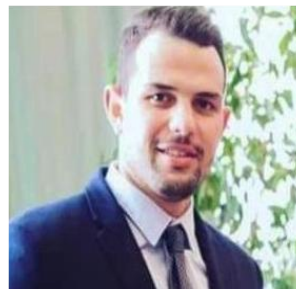
Sahar Zia received her PhD in Geography from the Department of Geography, University of the Punjab, Lahore, Pakistan in 2020. She is currently an Assistant Professor of Geography at the Geography Department, Lahore College for Women University (LCWU), Lahore, Pakistan. She has authored 30 publications in national and

international journals and has been a frequent presenter at conferences, contributing significantly to geography and urban informatics. With proficiency in advanced software such as ArcGIS Pro, QGIS, ENVI, ERDAS IMAGINE, and Google Earth Engine, she has conducted numerous research studies with GIS integration like LULC classification, time-series analysis of satellite imagery, geostatistical modeling, and scenario-based climate impact assessments. Dr. Zia has a strong research background in addressing global challenges such as deforestation, biodiversity loss, climate change, and pollution, alongside her ability to link human systems transformation to environmental outcomes.



M. Abdullah-Al-Wadud is working as an associate professor in the Department of Software Engineering at King Saud University, Saudi Arabia. He received his B.S. degree in computer science and M.S. in computer science and engineering from the University of Dhaka, Bangladesh, in 2003 and 2004, respectively. In 2009, he

completed his PhD in Computer Engineering from Kyung Hee University, South Korea. Afterward, he served as a faculty member of the Department of Industrial and Management Engineering at Hankuk University of Foreign Studies, South Korea, from 2009 to 2014. He also served as a lecturer at Daffodil International University, Bangladesh, and East West University, Bangladesh. His research interests include artificial intelligence, optimization, computer vision, cloud computing, recommender systems, software engineering, sensor and ad hoc networks.



Rebouh. N.Y. PhD in Agricultural Sciences, Associate Professor at the Department of Environmental Management, Institute of Environmental Engineering, RUDN University. Graduated from the Agrarian and Technological Institute of RUDN University, specializing in crop production. Author of more than

70 scientific papers indexed in international Scopus/WoS databases. Laureate of the prize of the best young scientist of RUDN University in 2024. His research interests include precision agriculture, soil contamination, Vis-NIR Spectroscopy, eco-farming systems, process modeling, phytopathology, entomology, geospatial data science, natural hazards, and environmental monitoring.